solution can be obtained in the large-discharge-current limit for the transient discharge current. Integration of the transient discharge current shows that 2/3 of the total stored charge in the narrow base diffuses back to the emitter when the base region is discharged. This fraction is the same as the fraction of total stored charge in the base contributing to the diffusion capacitance. In other words, one can think of the diffusion capacitance as coming from the portion of the stored minority charge that is "reclaimable" in the form of an ac current as the diode responds to an ac signal (Lindmayer and Wrigley, 1965).]

It is instructive to examine the relative magnitude of the two capacitance components C_{Dn} and C_{Dp}. Using the hole equivalent of Eq. (2.127) for hole current and Eq. (2.131) for electron current, and the relationship in Eq. (2.116), we have

$$\frac{C_{Dn}\,(\text{narrow base})}{C_{Dp}\,(\text{wide emitter})} = \frac{2}{3}\frac{N_E}{N_B}\frac{W_B}{L_{pE}}. \tag{2.168}$$

The ratio N_E/N_B is typically about 100 for an n^+–p diode. For an emitter with $N_E = 1 \times 10^{20}\text{cm}^{-3}$, L_{pE} is about 0.3 μm (see Fig. 2.24). Therefore, for practical one-sided diodes where the base width is larger than 0.03 μm, the ratio C_{Dn}/C_{Dp} is much larger than unity. That is, the diffusion capacitance of a one-sided p–n diode is dominated by the minority charge stored in the base of the diode. The diffusion capacitance due to the minority charge stored in the emitter is small in comparison. The effect of heavy doping, when included, will increase the amount of stored charge and hence the diffusion capacitance. Since the heavy-doping effect is larger in the more heavily doped emitter than in the base, it will make the ratio C_{Dn}/C_{Dp} smaller than that given by Eq. (2.168) (See Exercise 2.18).

2.3 MOS Capacitors

The metal–oxide–semiconductor (MOS) structure is the basis of CMOS technology. The Si–SiO_2 MOS system has been studied extensively (Nicollian and Brews, 1982) because it is directly related to most planar devices and integrated circuits. In this section, we review the fundamental properties of MOS capacitors and the basic equations that govern their operation. The effects of charges in the oxide layer and at the oxide–silicon interface are discussed in Section 2.3.6.

2.3.1 Surface Potential: Accumulation, Depletion, and Inversion

2.3.1.1 Energy-Band Diagram of an MOS System

The cross section of an MOS capacitor is shown in Fig. 2.27. It consists of a conducting gate electrode (metal or heavily doped polysilicon) on top of a thin layer of silicon dioxide grown on a silicon substrate. The energy band diagrams of the three components when separate are shown in Figure. 2.28. Before we discuss the energy band diagram of an MOS device, it is necessary to first introduce the concept of *free electron level* and *work function* which play key roles in the relative energy band placement when two different materials are brought into contact. Figure 2.28(c) shows the band diagram of a

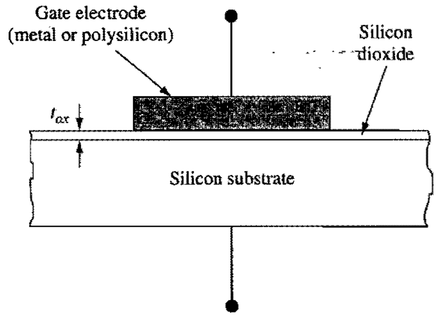

Figure 2.27. Schematic cross section of an MOS capacitor.

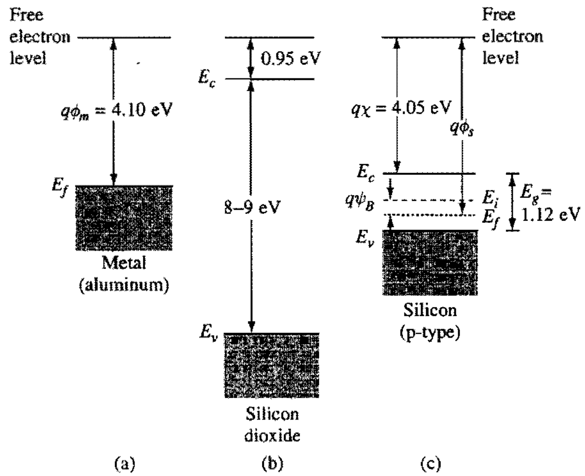

Figure 2.28. Energy-band diagram of the three components of an MOS capacitor: (a) metal (aluminum), (b) silicon dioxide, and (c) p-type silicon.

p-type silicon with the addition of the free electron level at some energy above the conduction band. The free electron level is defined as the energy level above which the electron is free, i.e., no longer bonded to the lattice.[3] In silicon, the free electron level is 4.05 eV above the conduction band edge, as shown in Figure. 2.28(c). In other words, an electron at the conduction band edge must gain an additional energy of 4.05 eV (called *the electron affinity*, $q\chi$) in order to break loose from the crystal field of silicon. Figure 2.28(b) shows the band diagram of silicon dioxide – an insulator with a large energy gap in the range of 8–9 eV. The free electron level in silicon dioxide is 0.95 eV above its conduction band.

[3] In other texts, the free electron level is often referred to as the *vacuum level*. Here we use a different term to avoid the implication that the vacuum level is universal.

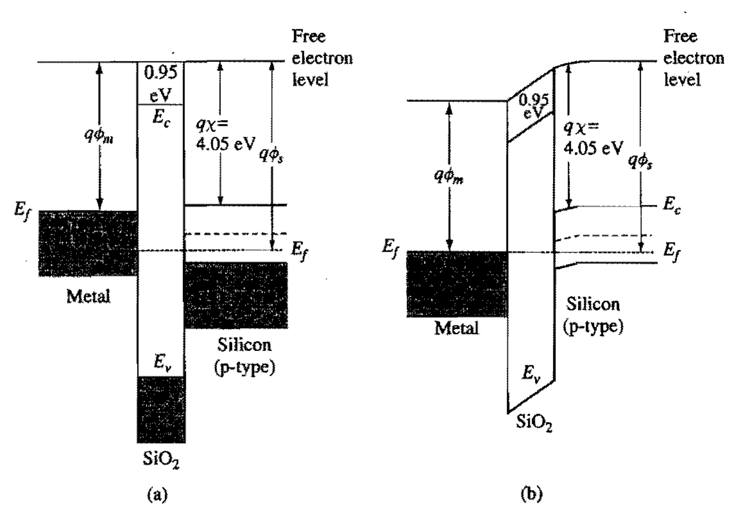

Figure 2.29. Band diagrams of an MOS system under (a) the flatband condition, and (b) zero gate-voltage condition.

Work function is defined as the energy difference between the free electron level and the Fermi level. For the p-type silicon example in Fig. 2.28(c), the work function, $q\phi_s$, can be expressed as:

$$q\phi_s = q\chi + \frac{E_g}{2} + q\psi_B. \qquad (2.169)$$

Here ψ_B is the difference between the Fermi potential and the intrinsic potential given by Eq. (2.48). The same definition of work function, $q\phi_m$, applies to metals (remember that the conduction band is half filled in metals), as shown in Fig. 2.28(a). It means that an electron at the Fermi level needs to receive an energy equal to $q\phi_m$ to be free from the metal. Different metals have different work functions.

When two different materials are brought into contact, they must share the same free electron level at the interface, i.e., the free electron level is continuous from one material to the next. This is because at the interface of two materials, an electron that is free from the crystal field of one material is also free from the crystal field of the other material. Figure 2.29(a) shows the band diagram of an MOS system under the *flatband condition* in which there is no field in all three materials. Since for this example the metal work function is less than the silicon work function, the flatband condition is reached by applying a negative gate voltage, $-(\phi_s - \phi_m) \equiv \phi_{ms}$, called the *flatband voltage*, with respect to the silicon substrate. This is seen in Fig. 2.29(a) as the displacement between the two Fermi levels. In general, the flatband voltage of an MOS device is given by

$$V_{fb} = (\phi_m - \phi_s) - \frac{Q_{ox}}{C_{ox}} = \phi_{ms} - \frac{Q_{ox}}{C_{ox}}, \qquad (2.170)$$

where Q_{ox} is the equivalent oxide charge per unit area at the oxide–silicon interface (defined in Section 2.3.7), and C_{ox} is the oxide capacitance per unit area,

$$C \quad \varepsilon_{ox} \qquad (2.171)$$

for an oxide film of thickness t_{ox} and permittivity ε_{ox}. In modern VLSI technologies, Q_{ox}/q at the Si–SiO_2 interface can be controlled to below 10^{10} cm^{-2} (positive) for $\langle 100 \rangle$-oriented surfaces. Its contribution to the flatband voltage is less than 50 mV for thin gate oxides used in 1-μm technology and below ($t_{ox} \leq 20$ nm). Therefore, the flatband voltage is mainly determined by the work function difference ϕ_{ms}.

Donde las bandas se curvan hay campo eléctrico
$E = -q\phi(x)$
$\frac{dE}{dx} = q \cdot \mathcal{E}(x)$
$\mathcal{E}(x) = \frac{1}{q} \cdot \frac{dE}{dx}$

At zero gate voltage when the Fermi levels line up, electric fields are developed in both the oxide and silicon, as shown in Fig. 2.29(b). One should note that the electron affinity, $q\chi$, is a material property which depends only on the type of the semiconductor and does not change with either the location or the doping type. When there is band bending as in Fig. 2.29(b) or, e.g., as in the depletion region of a p–n junction, the free electron level would bend in parallel with the conduction band such that the distance between the free electron level and the conduction band remains constant, much like the energy gap. While in this case the free electron level is not flat within one material, it must still be continuous between adjacent materials. Because of the common free electron level at the interface, ***the electron energy barrier is $q\phi_{ox} = 4.05\,eV - 0.95\,eV = 3.1\,eV$ between the conduction bands of silicon and silicon dioxide***. This figure has important significance when discussing the reliability of Si–SiO_2 systems (Section 2.5.3). For simplicity, the free electron level is mostly omitted in subsequent MOS band diagrams. One should keep in mind its essential role in setting the relative band placement between different materials.

The same principle of continuity of free-electron level at interface applies to metal–semiconductor junctions (Schottky diodes discussed in Section 2.4.1) and heterojunction devices with different bandgaps in establishing their band alignment relationships.

2.3.1.2 Gate Voltage and Surface Potential

Figure 2.30 shows the band diagrams of an MOS device when different gate voltages are applied. Free electron levels are omitted. The top level in the oxide region represents the

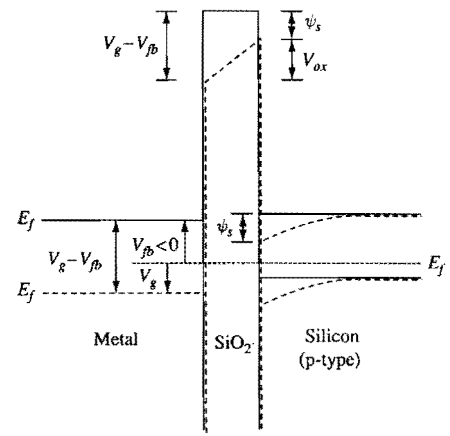

Figure 2.30. Energy band and potential diagrams of an MOS capacitor showing how the bands change under different gate bias conditions. The solid lines represent the flatband condition. The dashed lines represent the condition when a positive gate voltage is applied. Note that in the diagram the electron

conduction band. The same energy band diagrams are also used as potential diagrams to show the relationship between the applied voltage and the potential drop in different regions of the device. The applied gate voltage is referenced with respect to the Fermi level of the p-type substrate. In the flatband condition depicted by the solid lines in Fig. 2.30, $V_{fb} < 0$ is applied to the gate, as in the case of Fig. 2.29(a). When a positive gate voltage, V_g, is applied, as depicted by the dashed lines in Fig. 2.30, the metal Fermi level is displaced downward from that of the flatband condition by a total amount of $V_g - V_{fb}$. Because of the fixed band relationship between the metal and oxide, the oxide conduction band on the metal side is also displaced downward by the same amount. This causes a field to develop in the oxide and, at the same time, a downward bending of the bands in the p-type silicon near the surface. The amount of band bending in silicon is defined as the *surface potential*, ψ_s, i.e., the potential at the silicon surface relative to that in the bulk substrate. Because of the fixed band relationship between the oxide and silicon, it is clear that

Como V_g impacta em ψ_s

$$V_g - V_{fb} = \psi_s + V_{ox}, \tag{2.172}$$

where V_{ox} is the potential drop across the oxide, as indicated in Fig. 2.30.

How $V_g - V_{fb}$ is partitioned into ψ_s and V_{ox} depends on both the oxide thickness and the doping concentration of the p-type silicon. Based on the dielectric boundary conditions discussed in Section 2.1.4.2, a field relationship exists at the silicon–oxide interface,

$$\varepsilon_{ox}\mathscr{E}_{ox} = \varepsilon_{si}\mathscr{E}_s, \tag{2.173}$$

→ conforme

or $\mathscr{E}_{ox} \approx 3\mathscr{E}_s$, assuming negligible trapped charge at the interface. Note that the above equation applies to both the magnitude and the direction of the fields. In most cases, there is negligible net charge in the oxide and Poisson's equation becomes $d\mathscr{E}/dx = 0$. Therefore, the field in the oxide is constant,[4] and $V_{ox} = \mathscr{E}_{ox}t_{ox}$.

2.3.1.3 Accumulation, Depletion, and Inversion

Figure 2.31 shows the band diagrams of p-type ((a)–(d)) and n-type ((e)–(h)) MOS capacitors under different gate bias voltages with respect to the flatband voltage. For simplicity, the flatband voltage is taken to be zero for all cases. The flatband condition for p-type MOS discussed before is shown in Fig. 2.31(a). There is no charge, no field, and the carrier concentration equals the ionized acceptor concentration throughout the silicon. Now consider the case when a negative voltage is applied to the gate of a p-type MOS capacitor, as shown in Fig. 2.31(b). This raises the metal Fermi level (i.e., electron energy) with respect to the silicon Fermi level and creates an electric field in the oxide that would accelerate a negative charge toward the silicon substrate. A field is also induced at the silicon surface in the same direction as the oxide field. Because of the low carrier concentration in silicon (compared with metal), the bands bend upward toward the oxide interface. ***The Fermi level stays flat within the silicon, since there is no net flow of conduction current,*** as was discussed in Section 2.1.4.5. Due to the band bending, the

[4] If the field in the oxide is not constant, $\mathscr{E}_{ox}$ in Eq. (2.173) is defined as the oxide field at the oxide–silicon interface.

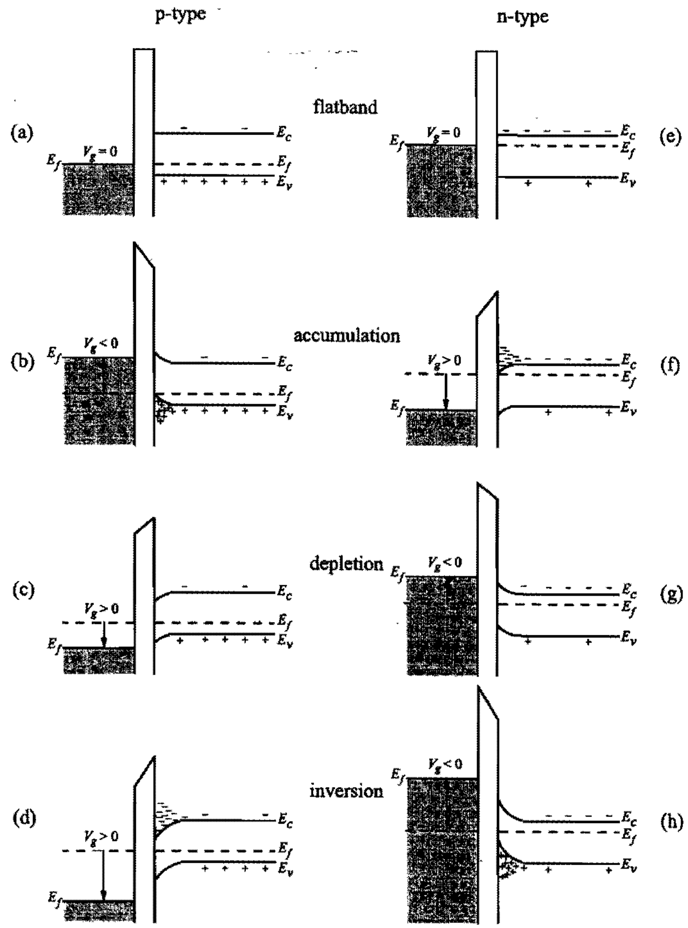

Figure 2.31. Energy-band diagrams for ideal (zero flatband voltage) (a)–(d) p-type and (e)–(h) n-type MOS capacitors under different bias conditions: (a), (e), flat band; (b), (f), accumulation; (c), (g), depletion; (d), (h), inversion. (After Sze, 1981.)

valence band at the surface is much closer to the Fermi level than is the valence band in the bulk silicon. This results in a hole concentration much higher at the surface than the equilibrium hole concentration in the bulk. Since excess holes are accumulated at the surface, this is referred to as the *accumulation* condition. One can think of the excess holes as being attracted toward the surface by the negative gate voltage. An equal amount of negative charge appears on the metal side of the MOS capacitor, as required for charge neutrality.

On the other hand, if a positive voltage is applied to the gate of a p-type MOS capacitor, the metal Fermi level moves downward, which creates an oxide field in the direction of accelerating a negative charge toward the metal electrode. A similar field is induced in the silicon, which causes the bands to bend downward toward the surface, as shown in Fig. 2.31(c). Since the valence band at the surface is now farther away from the Fermi level than is the valence band in the bulk, the hole concentration at the surface is lower than the concentration in the bulk. This is referred to as the *depletion* condition. One can think of the holes as being repelled away from the surface by the positive gate voltage. The situation is similar to the depletion layer in a p–n junction discussed in Section 2.2.2. The depletion of holes at the surface leaves the region with a net negative charge arising from the unbalanced acceptor ions. An equal amount of positive charge appears on the metal side of the capacitor.

As the positive gate voltage increases, the band bending also increases, resulting in a wider depletion region and more (negative) depletion charge. This goes on until the bands bend downward so much that at the surface, the conduction band is closer to the Fermi level than the valence band is, as shown in Fig. 2.31(d). When this happens, not only are the holes depleted from the surface, but the surface potential is such that it is energetically favorable for electrons to populate the conduction band. In other words, the surface behaves like n-type material with an electron concentration given by Eq. (2.49). ***Note that this n-type surface is formed not by doping, but instead by inverting the original p-type substrate with an applied electric field.*** This condition is called *inversion*. The negative charge in the silicon consists of both the ionized acceptors and the thermally generated electrons in the conduction band. Again, it is balanced by an equal amount of positive charge on the metal gate. The surface is inverted as soon as $E_i = (E_c + E_v)/2$ crosses E_f. This is called *weak inversion* because the electron concentration remains small until E_i is considerably below E_f. If the gate voltage is increased further, the concentration of electrons at the surface will be equal to, and then exceed, the hole concentration in the substrate. This condition is called *strong inversion*.

So far we have discussed the band bending for accumulation, depletion, and inversion of silicon surface in a p-type MOS capacitor. Similar conditions hold true in an n-type MOS capacitor, except that the polarities of voltage, charge, and band bending are reversed, and the roles of electrons and holes are interchanged. The band diagrams for flatband, accumulation, depletion, and inversion conditions of an n-type MOS capacitor are shown in Fig. 2.31(e)–(h), where the metal work function per electron charge ϕ_m is assumed to be equal to that of the n-type silicon, given by

$$\phi_s = \chi + \frac{E_g}{2q} - \psi_B, \tag{2.174}$$

instead of Eq. (2.169). Accumulation occurs when a positive voltage is applied to the metal gate and the silicon bands bend downward at the surface. Depletion and inversion occur when the gate voltage is negative and the bands bend upward toward the surface.

2.3.2 Electrostatic Potential and Charge Distribution in Silicon

2.3.2.1 Solving Poisson's Equation → todo sur le Poisson

In this sub-subsection, the relations among the surface potential, charge, and electric field are derived by solving Poisson's equation in the surface region of silicon. A more detailed band diagram at the surface of a p-type silicon is shown in Fig. 2.32. The potential $\psi(x)$ =

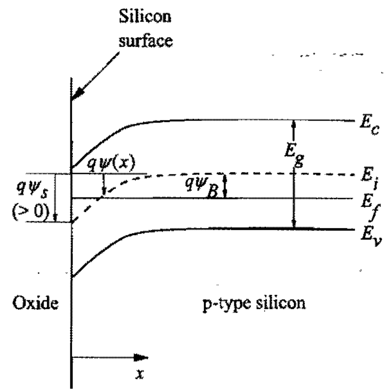

Figure 2.32. Energy-band diagram near the silicon surface of a p-type MOS device. The band bending ψ is defined as positive when the bands bend downward with respect to the bulk. Accumulation occurs when $\psi_s < 0$. Depletion and inversion occur when $\psi_s > 0$.

$\psi_i(x) - \psi_i(x=\infty)$ is defined as the amount of band bending at position x, where $x=0$ is at the silicon surface and $\psi_i(x=\infty)$ is the intrinsic potential in the bulk silicon. Remember that $\psi(x)$ is positive when the bands bend downward. The boundary conditions are $\psi=0$ in the bulk silicon, and $\psi=\psi(0)=\psi_s$ at the surface. The surface potential ψ_s depends on the applied gate voltage, as discussed in Section 2.3.1.2. Poisson's equation, Eq. (2.44), is

$$\frac{d^2\psi}{dx^2} = -\frac{d\mathscr{E}}{dx} = -\frac{q}{\varepsilon_{si}}\left[p(x) - n(x) + N_d^+(x) - N_a^-(x)\right]. \tag{2.175}$$

For a uniformly doped p-type substrate of acceptor concentration N_a with complete ionization, $N_d^+(x) - N_a^-(x) = -N_a$, independent of x. Charge neutrality condition deep in the bulk substrate requires

$$N_d^+(x) - N_a^-(x) = -N_a = -p_0 + \frac{n_i^2}{p_0}, \tag{2.176}$$

where $p_0 = p(x=\infty)$ and $n_i^2/p_0 = n(x=\infty)$ are the majority (holes) and minority (electrons) carrier densities in the bulk substrate, respectively. In general, $p(x)$ and $n(x)$ are given by Eq. (2.50) and Eq. (2.49), which can be expressed in terms of $\psi(x)$ using $\psi(x) = \psi_i(x) - \psi_i(x=\infty)$ and $\psi_B = \psi_f - \psi_i(x=\infty)$ as defined in Fig. 2.32:

$$p(x) = n_i e^{q[\psi_f - \psi_i(x)]/kT} = n_i e^{q[\psi_B - \psi(x)]/kT} = p_0\, e^{-q\psi(x)/kT} \tag{2.177}$$

and

$$n(x) = n_i e^{q[\psi_i(x) - \psi_f]/kT} = n_i e^{q[\psi(x) - \psi_B]/kT} = \frac{n_i^2}{p_0}\, e^{q\psi(x)/kT}. \tag{2.178}$$

Note that ψ_f is independent of x because there is no net current flow perpendicular to the surface and the Fermi level stays flat. Also note that $p_0 = p(x=\infty) = n_i \exp(q\psi_B/kT)$ and $n_i^2/p_0 = n(x=\infty) = n_i \exp(-q\psi_B/kT)$ in the last step of the above equations.

In practice, $N_a \gg n_i$, and $p_0 \approx N_a$ from Eq. (2.176). Substituting the last three equations into Eq. (2.175) and replacing p_0 by N_a yield

$$\frac{d^2\psi}{dx^2} = -\frac{q}{\varepsilon_{si}}\left[N_a\left(e^{-q\psi/kT}-1\right) - \frac{n_i^2}{N_a}\left(e^{q\psi/kT}-1\right)\right]. \tag{2.179}$$

Multiplying $(d\psi/dx)\,dx$ on both sides of Eq. (2.179) and integrating from the bulk ($\psi = 0$, $d\psi/dx = 0$) toward the surface, one obtains

$$\int_0^{d\psi/dx} \frac{d\psi}{dx}\, d\left(\frac{d\psi}{dx}\right) = -\frac{q}{\varepsilon_{si}}\int_0^{\psi}\left[N_a\left(e^{-q\psi/kT}-1\right) - \frac{n_i^2}{N_a}\left(e^{q\psi/kT}-1\right)\right]d\psi, \tag{2.180}$$

which gives the electric field at x, $\mathscr{E} = -d\psi/dx$, in terms of ψ:

$$\mathscr{E}^2(x) = \left(\frac{d\psi}{dx}\right)^2 = \frac{2kTN_a}{\varepsilon_{si}}\left[\left(e^{-q\psi/kT} + \frac{q\psi}{kT} - 1\right) + \frac{n_i^2}{N_a^2}\left(e^{q\psi/kT} - \frac{q\psi}{kT} - 1\right)\right]. \tag{2.181}$$

At $x = 0$, we let $\psi = \psi_s$ and $\mathscr{E} = \mathscr{E}_s$. From Gauss's law, Eq. (2.43), the total charge per unit area induced in the silicon (equal and opposite to the charge on the metal gate) is

Carga TOTAL

$$Q_s = -\varepsilon_{si}\mathscr{E}_s = \pm\sqrt{2\varepsilon_{si}kTN_a}\left[\left(e^{-q\psi_s/kT} + \frac{q\psi_s}{kT} - 1\right) + \frac{n_i^2}{N_a^2}\left(e^{q\psi_s/kT} - \frac{q\psi_s}{kT} - 1\right)\right]^{1/2}. \tag{2.182}$$

This function is plotted in Fig. 2.33. At the flat-band condition, $\psi_s = 0$ and $Q_s = 0$. In accumulation, $\psi_s < 0$ (bands bending upward) and the first term in the square brackets dominates once $-q\psi_s/kT > 1$. The accumulation charge density is then proportional to $\exp(-q\psi_s/2kT)$ as indicated in the Fig 2.33.[5] In depletion, $\psi_s > 0$ and $q\psi_s/kT > 1$, but $\exp(q\psi_s/kT)$ is not large enough to make the n_i^2/N_a^2 term appreciable. Therefore, the $q\psi_s/kT$ term in the square brackets dominates and the negative depletion charge density (from ionized acceptor atoms) is proportional to $\psi_s^{1/2}$. When ψ_s increases further, the $(n_i^2/N_a^2)\exp(q\psi_s/kT)$ term eventually becomes larger than the $q\psi_s/kT$ term and dominates the square bracket. This is when inversion occurs. The negative inversion charge density is proportional to $\exp(q\psi_s/2kT)$ as indicated in Fig. 2.33.

explicación de cada tramo del gráfico

A popular criterion for the onset of strong inversion is for the surface potential to reach a value such that $(n_i^2/N_a^2)\exp(q\psi_s/kT) = 1$, i.e.,

cuándo empieza la inversión

$$\psi_s(\text{inv}) = 2\psi_B = 2\frac{kT}{q}\ln\left(\frac{N_a}{n_i}\right). \tag{2.183}$$

[5] These results stem from the Maxwell–Boltzmann approximation made in Section 2.1.1.3, which over-estimates the occupancy of electron states near and below the Fermi level. For accumulation and inversion layers with carrier densities in the degenerate range, i.e., when $\psi_s < -0.2$ V or > 0.9 V where the Fermi level goes into the valence or the conduction band, the more exact Fermi–Dirac distribution gives a less steep rise of the sheet charge density with the surface potential.

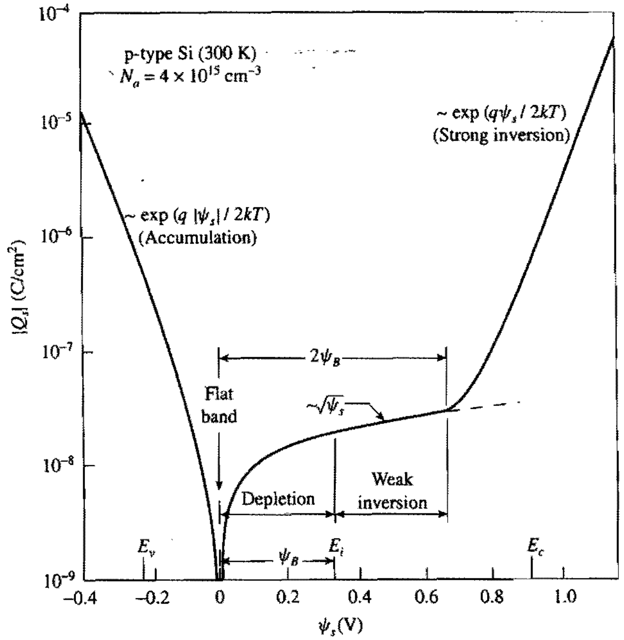

Figure 2.33. Variation of total charge density (fixed plus mobile) in silicon as a function of surface potential ψ_s for a p-type MOS device. The labels E_v, E_i, E_c indicate the surface potential values where the valence band, the intrinsic level, and the conduction band cross the Fermi level. (After Sze, 1981.)

→ martín habra dicho algo de esto?
Espero que no

Under this condition, the electron concentration given by Eq. (2.178) at the surface becomes equal to the depletion charge density N_a. ***After inversion takes place, even a slight increase in the surface potential results in a large buildup of electron density at the surface.*** The inversion layer effectively shields the silicon from further penetration of the gate field. Since almost all of the incremental charge is taken up by electrons, there is no further increase of either the depletion charge or the depletion-layer width. The expression in Eq. (2.183) is a rather weak function of the substrate doping concentration. For typical values of $N_a = 10^{16}$–10^{18} cm^{-3}, $2\psi_B$ varies only slightly, from 0.70 to 0.94 V.

2.3.2.2 Depletion Approximation

In general, Eq. (2.181) must be solved numerically to obtain $\psi(x)$. In particular cases, approximations can be made to allow the integral to be carried out analytically. For example, in the depletion region where $2\psi_B > \psi > kT/q$, only the $q\psi/kT$ term in the square bracket needs to be kept and

$$\frac{d\psi}{dx} = -\sqrt{\frac{2qN_a\psi}{\varepsilon_{si}}}. \tag{2.184}$$

One can then rearrange the factors and integrate:

$$\int_{\psi_s}^{\psi} \frac{d\psi}{\sqrt{\psi}} = -\int_0^x \sqrt{\frac{2qN_a}{\varepsilon_{si}}}\, dx, \tag{2.185}$$

where ψ_s is the surface potential at $x=0$ as assumed before. Therefore,

$$\psi = \psi_s \left(1 - \sqrt{\frac{qN_a}{2\varepsilon_{si}\psi_s}}\, x\right)^2, \tag{2.186}$$

which can be written as

$$\psi = \psi_s \left(1 - \frac{x}{W_d}\right)^2. \tag{2.187}$$

This is a parabolic equation with the vertex at $\psi=0$, $x=W_d$, where

$$W_d = \sqrt{\frac{2\varepsilon_{si}\psi_s}{qN_a}} \tag{2.188}$$

is the depletion-layer width defined as the distance to which the band bending extends. The total depletion charge density in silicon, Q_d, is equal to the charge per unit area of ionized acceptors in the depletion region: ↳ Carga de vaciamiento

$$Q_d = -qN_aW_d = -\sqrt{2\varepsilon_{si}qN_a\psi_s}. \tag{2.189}$$

These results are very similar to those of the one-sided abrupt p–n junction under the depletion approximation, discussed in Section 2.2.2. ***In the MOS case, however, W_d reaches a maximum value W_{dm} at the onset of strong inversion when $\psi_s = 2\psi_B$.*** Substituting Eq. (2.183) into Eq. (2.188) gives the maximum depletion width:

$$W_{dm} = \sqrt{\frac{4\varepsilon_{si}kT\ln(N_a/n_i)}{q^2N_a}}. \tag{2.190}$$

2.3.2.3 Strong Inversion

Beyond strong inversion, the $(n_i^2/N_a^2)\exp(q\psi/kT)$ term representing the inversion charge in Eq. (2.181) becomes appreciable and must be kept, together with the depletion charge term:

$$\frac{d\psi}{dx} = -\sqrt{\frac{2kTN_a}{\varepsilon_{si}}\left(\frac{q\psi}{kT} + \frac{n_i^2}{N_a^2}e^{q\psi/kT}\right)}. \tag{2.191}$$

This equation can only be integrated numerically. The boundary condition is $\psi = \psi_s$ at $x=0$. After $\psi(x)$ is solved, the electron distribution $n(x)$ in the inversion layer can be calculated from Eq. (2.178). Examples of numerically calculated $n(x)$ are plotted in Fig. 2.34 for two values of ψ_s with $N_a = 10^{16}$ cm^{-3}. ***The electrons are distributed extremely close to the surface with an inversion-layer width less than 50 Å***. A higher surface potential or field tends to confine the electrons even closer to the surface. In

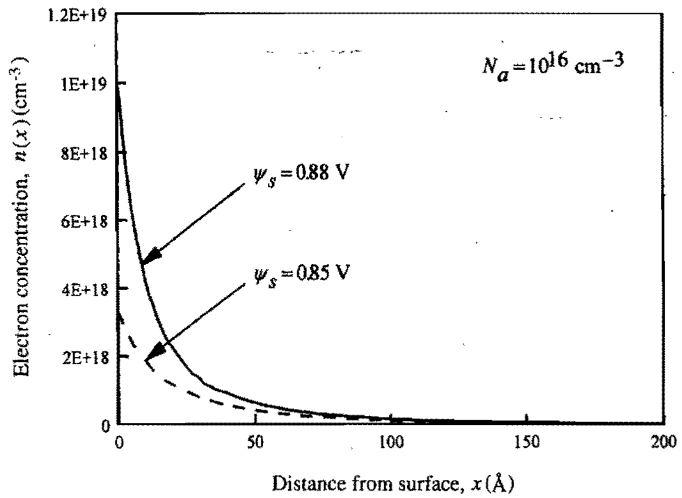

Figure 2.34. Electron concentration versus distance in the inversion layer of a p-type MOS device.

general, electrons in the inversion layer must be treated quantum-mechanically as a 2-D gas (Stern and Howard, 1967). According to the quantum-mechanical model, inversion-layer electrons occupy discrete energy bands and have a peak distribution 10–20 Å away from the surface. More details will be discussed in Section 4.2.4.

When the inversion charge density per unit area, Q_i, is much greater than the depletion charge density, Eq. (2.182) can be approximated by ↳ Carga de inversión

$$Q_i = -\sqrt{\frac{2\varepsilon_{si}kTn_i^2}{N_a}}\, e^{q\psi_s/2kT}. \tag{2.192}$$

Since the electron concentration at the surface is

$$n(0) = \frac{n_i^2}{N_a} e^{q\psi_s/kT}, \tag{2.193}$$

one can write

$$|Q_i| = \sqrt{2\varepsilon_{si}kTn(0)}. \tag{2.194}$$

The effective inversion-layer thickness (classical model) can be estimated from $Q_i/qn(0) = 2\varepsilon_{si}kT/qQ_i$, which is inversely proportional to Q_i. Similar expressions also hold true for the surface charge density of extra holes under accumulation, except that the factor n_i^2/N_a is replaced by N_a.

2.3.2.4 Surface Potential and Charge Density as a Function of Gate Voltage

In Section 2.3.2.1, charge and potential distributions in silicon were solved in terms of the surface potential ψ_s as a boundary condition. ψ_s is not directly measurable, but is controlled by and can be determined from the applied gate voltage. The gate voltage

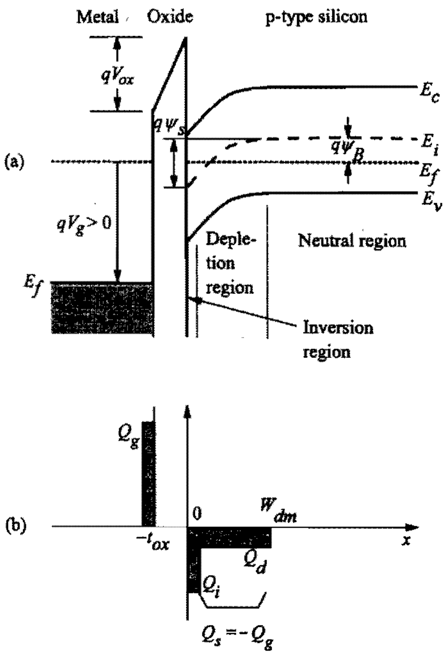

Figure 2.35. (a) Band diagram of a p-type MOS capacitor with a positive voltage applied to the gate ($V_{fb} = 0$). (b) Charge distribution under inversion condition.

equation, Eq. (2.172), relates the potential drop V_{ox} across the oxide and the band bending ψ_s in silicon to the departure from the flatband condition due to the applied gate voltage V_g (Fig. 2.35(a)). Assuming negligible fixed charges in the oxide, the potential drop V_{ox} can be expressed as $\mathscr{E}_{ox}t_{ox}$, which equals $(\varepsilon_{si}/\varepsilon_{ox})\mathscr{E}_s t_{ox}$ based on the boundary condition, Eq. (2.173). Applying Gauss's law, $Q_s = -\varepsilon_{si}\mathscr{E}_s$, ***the gate bias equation becomes***

V_g tiene q' deshacer V_{FB}, cargar el capacitor y doblar las bandas ←

$$V_g - V_{fb} = V_{ox} + \psi_s = \frac{-Q_s}{C_{ox}} + \psi_s, \tag{2.195}$$

where Q_s is the total charge per unit area induced in the silicon, and $C_{ox} = \varepsilon_{ox}/t_{ox}$ is the oxide capacitance per unit area for an oxide of thickness t_{ox}. There is a negative sign in front of Q_s in Eq. (2.195) because the charge on the metal gate is always equal but opposite to the charge in silicon, i.e., Q_s is negative when $V_g - V_{fb}$ is positive and vice versa. The charge distribution in an MOS capacitor is shown schematically in Fig. 2.35 (b), where the total charge Q_s may include both depletion and inversion components. For simplicity of discussion, oxide and interface trapped charges are ignored here. They will be discussed in detail in Sections 2.3.6 and 2.3.7.

In general, Q_s is a function of ψ_s given by Eq. (2.182), and plotted in Fig. 2.33. Equation (2.195) is then an implicit equation that can be solved for ψ_s as a function of V_g. An example of the numerical solution is shown in Fig. 2.36. Below the condition for strong inversion, $\psi_s = 2\psi_B$, ψ_s increases more or less linearly with V_g. Beyond $\psi_s = 2\psi_B$, ψ_s nearly saturates –

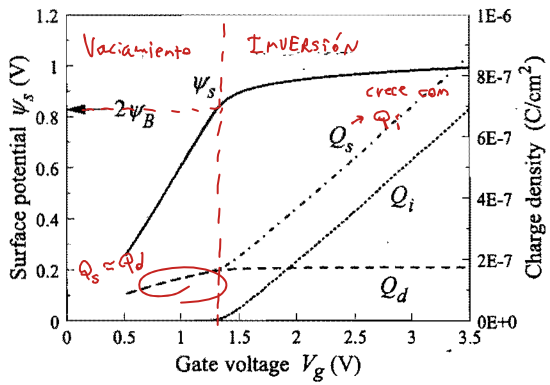

Figure 2.36. Numerical solutions of surface potential, total silicon charge density, inversion charge density, and depletion charge density from the gate voltage equation (2.195) coupled to Eq. (2.182). The MOS device parameters are $N_a = 10^{17}$ cm^{-3}, $t_{ox} = 10$ nm, and $V_{fb} = 0$.

explicación del gráfico

increasing by less than 0.2 V while V_g increases by 2 V. After ψ_s is solved, Q_s is calculated and plotted as a function of V_g in Fig. 2.36. By numerically evaluating the integrals in Exercise 2.6, Q_s is separated into its two components, the depletion charge density Q_d and the inversion charge density Q_i, which are also plotted in Fig. 2.36. ***It is clear that before the $\psi_s = 2\psi_B$ condition, the charge in the silicon is predominantly of the depletion type.*** Under such depletion conditions, $Q_s(\psi_s) = Q_d(\psi_s)$, an analytical expression for $\psi_s(V_g)$ can be derived by solving a quadratic equation (see Eq. (2.202)). After $\psi_s = 2\psi_B$, the depletion charge no longer increases with V_g because of shielding by the inversion layer discussed before. ***Almost all of the increase of Q_s beyond $\psi_s = 2\psi_B$ is taken up by Q_i with a slope $dQ_i/dV_g \approx C_{ox}$.*** While on the linear scale it appears that Q_i is zero below the $\psi_s = 2\psi_B$ threshold, on the log scale it can be seen that Q_i actually remains finite and decreases exponentially with V_g. It is the source of the subthreshold leakage current in MOSFETs – an important design consideration further addressed in detail in Section 3.1.3.2. Under extreme accumulation and inversion conditions, $-Q_s \approx C_{ox}(V_g - V_{fb})$, since both V_g and V_{ox} can be much larger than the silicon bandgap, $E_g/q = 1.12$ V (for CMOS technologies with $V_{dd} \gg 1$ V), while ψ_s is at most comparable to E_g/q (surface potential pinned to either the valence band or the conduction band edge).

2.3.3 Capacitances in an MOS Structure

2.3.3.1 Definition of Small-Signal Capacitances

We now consider the capacitances in an MOS structure. In most cases, MOS capacitances are defined as small-signal differential of charge with respect to voltage or potential. They can easily be measured by applying a small ac voltage on top of a dc bias across the device and sensing the out-of-phase ac current at the same frequency (the in-phase component gives the small-signal conductance). The total MOS capacitance per unit area is

$$C_g = \frac{d(-Q_s)}{dV_g}. \tag{2.196}$$

If we differentiate Eq. (2.195) with respect to $-Q_s$ and define the silicon part of the capacitance as

$$C_{si} = \frac{d(-Q_s)}{d\psi_s}, \tag{2.197}$$

capacitores en serie

we obtain

$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{d\psi_s}{d(-Q_s)} = \frac{1}{C_{ox}} + \frac{1}{C_{si}}. \tag{2.198}$$

In other words, the total capacitance equals the oxide capacitance and the silicon capacitance connected in series. The capacitances are defined in such a way that they are all positive quantities. An equivalent circuit is shown in Fig. 2.37(a). In reality, there is also an interface trap capacitance in parallel with C_{si}. It arises from charging and discharging of Si–SiO$_2$ interface traps and will be discussed in more detail in Section 2.3.7.

2.3.3.2 Capacitance–Voltage Characteristics: Accumulation

A typical capacitance-versus-gate-voltage (C–V) curve of a p-type MOS capacitor is plotted in Fig. 2.38, assuming zero flatband voltage. In fact, there are several different curves, depending on the frequency of the applied ac signal. We start with the "low-frequency" or *quasistatic* C–V curve. When the gate voltage is negative (by more than a few kT/q) with respect to the flatband voltage, the p-type MOS capacitor is in accumulation and $Q_s \propto \exp(-q\psi_s/2kT)$, as shown in Fig. 2.33. Therefore, $C_{si} = -dQ_s/d\psi_s = (q/2kT)Q_s = (q/2kT)C_{ox}|V_g - V_{fb} - \psi_s|$, and the MOS capacitance per unit area is given by

por esto era importante la fórmula de Q_s

$$\frac{1}{C_g} = \frac{1}{C_{ox}}\left[1 + \frac{2kT/q}{|V_g - V_{fb} - \psi_s|}\right]. \tag{2.199}$$

Since $2kT/q \approx 0.052$ V and ψ_s is limited to 0.1 to 0.3 V in accumulation, the MOS capacitance rapidly approaches C_{ox} when the gate voltage is 1–2 V more negative than the flat-band voltage.[6]

2.3.3.3 Capacitance at Flat Band

When the gate bias is zero in Fig. 2.38, the MOS is near the flat-band condition; therefore, $q\psi_s/kT \ll 1$. The inversion charge term in Eq. (2.182) can be neglected and the first exponential term can be expanded into a power series. Keeping only the first three terms of the series, one obtains $Q_s = -(\varepsilon_{si}q^2N_a/kT)^{1/2}\psi_s$. From Eq. (2.198), the flatband capacitance per unit area is given by

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$$\frac{1}{C_{fb}} = \frac{1}{C_{ox}} + \sqrt{\frac{kT}{\varepsilon_{si}q^2N_a}} = \frac{1}{C_{ox}} + \frac{L_D}{\varepsilon_{si}}, \tag{2.200}$$

[6] Actually, C_g approaches C_{ox} slower than that depicted by Eq. (2.199) because of the Fermi–Dirac distribution at degenerate carrier densities.

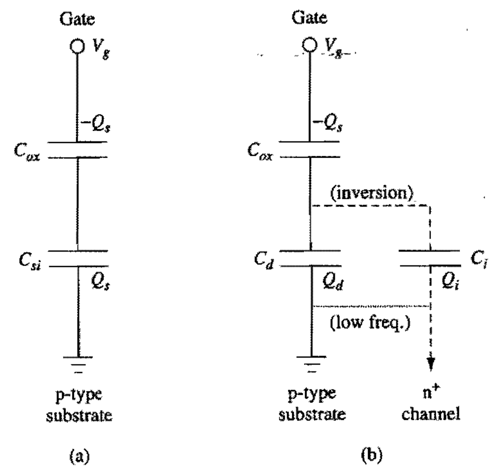

Figure 2.37. Equivalent circuits of an MOS capacitor. (a) All the silicon capacitances are lumped into C_{si}. (b) C_{si} is broken up into a depletion charge capacitance C_d and an inversion-layer capacitance C_i. C_d arises from the majority carriers, which can respond to high-frequency as well as low-frequency signals. C_i arises from the minority carriers, which can only respond to low-frequency signals, unless the surface inversion channel is connected to a reservoir of minority carriers as in a gated diode configuration. The thin dotted connection in (b) is effective only at low frequencies where minority carriers can respond.

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where L_D is the Debye length defined in Eq. (2.53). In most cases, C_{fb} is somewhat less than C_{ox}. For very thin oxides and low substrate doping, C_{fb} can be much smaller than C_{ox}.

2.3.3.4 Capacitance–Voltage Characteristics: Depletion

When the gate voltage is slightly higher than the flatband voltage in a p-type MOS capacitor, the surface starts to be depleted of holes; $1/C_{si}$ becomes appreciable and the capacitance decreases. Using the depletion approximation, one can find an analytical expression for C_g in this case. From Eq. (2.188) and Eq. (2.189),

$$C_d = \frac{d(-Q_d)}{d\psi_s} = \sqrt{\frac{\varepsilon_{si}qN_a}{2\psi_s}} = \frac{\varepsilon_{si}}{W_d}. \tag{2.201}$$

The last expression is identical to the depletion-layer capacitance per unit area in the p–n junction case discussed in Section 2.2.2. The bias equation (2.195) becomes

$$V_g - V_{fb} = \frac{qN_aW_d}{C_{ox}} + \psi_s = \frac{\sqrt{2\varepsilon_{si}qN_a\psi_s}}{C_{ox}} + \psi_s. \tag{2.202}$$

Substituting C_d from Eq. (2.201) for C_{si} in Eq. (2.198), and eliminating ψ_s using Eq. (2.202), one obtains

$$C_g = \frac{C_{ox}}{\sqrt{1 + [2C_{ox}^2(V_g - V_{fb})/(\varepsilon_{si}qN_a)]}}. \tag{2.203}$$

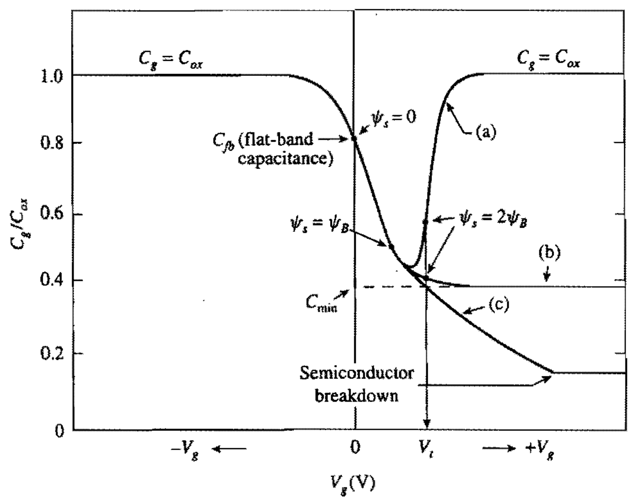


Figure 2.38. MOS capacitance-voltage curves: (a) low frequency, (b) high frequency, (c) deep depletion. $V_{fb} = 0$ is assumed. (After Size, 1981.)

This equation shows how the MOS capacitance decreases with increasing V_g under the depletion condition. It serves as a good approximation to the middle portions of the C–V curves in Fig. 2.38, provided that the MOS capacitor is not biased near the flat-band or the inversion condition.

2.3.3.5 Low-Frequency C–V Characteristics: Inversion

As the gate voltage increases further, however, the capacitance stops decreasing when $\psi_s = 2\psi_B$ [Eq. (2.183)] is reached and inversion occurs. ***Once the inversion layer forms, the capacitance starts to increase, since C_{si} is now given by the variation of the inversion charge with respect to ψ_s, which is much larger than the depletion capacitance.*** Assuming that the silicon charge is dominated by the inversion charge, one can carry out an approximation as in the accumulation case and show that the MOS capacitance in strong inversion is also given by Eq. (2.199). One difference is that ψ_s at inversion is in the range of 0.7 to 1.0 V, significantly higher than that at accumulation. In any case, the capacitance rapidly increases back to C_{ox} when the gate voltage is more than 2 to 3 V beyond the flat-band voltage, as shown in the low-frequency C–V curve (a) in Fig. 2.38.

2.3.3.6 High-Frequency Capacitance–Voltage Characteristics

The above discussion of the low-frequency MOS capacitance assumes that the minority carrier, i.e., the inversion charge, is able to follow the applied ac signal. This is true only if the frequency of the applied signal is lower than the reciprocal of the minority-carrier response time. The minority-carrier response time can be estimated from the generation–recombination current density, $J_R = qn_iW_d/\tau$, where τ is the minority-carrier lifetime discussed in Section 2.1.4. The time it takes to generate enough minority carriers to

replace something comparable to the depletion charge, $Q_d = qN_aW_d$, is on the order of $Q_d/J_R = (N_a/n_i)\tau$ (Jund and Poirier, 1966). This is typically 0.1–10 s. Therefore, ***for frequencies higher than 100 Hz or so, the inversion charge cannot respond to the applied ac signal.*** Only the depletion charge (majority carriers) can respond to the signal, which means that the silicon capacitance is given by C_d of Eq. (2.201) with W_d equal to its maximum value, W_{dm}, in Eq. (2.190). The high-frequency capacitance per unit area thus approaches a constant minimum value, C_{min}, at inversion given by

$$\frac{1}{C_{min}} = \frac{1}{C_{ox}} + \sqrt{\frac{4kT\ln(N_a/n_i)}{\varepsilon_{si}q^2N_a}}. \tag{2.204}$$

This is shown in the high-frequency C–V curve (b) in Fig. 2.38.

Typically, C–V curves are traced by applying a slow-varying ramp voltage to the gate with a small ac signal superimposed on it. However, if the ramp rate is fast enough that the ramping time is shorter than the minority-carrier response time, then there is insufficient time for the inversion layer to form, and the MOS capacitor is biased into deep depletion as shown by curve (c) in Fig. 2.38. In this case, the depletion width can exceed the maximum value given by Eq. (2.190), and the MOS capacitance decreases further below C_{min} until impact ionization takes place (Sze, 1981). Note that deep depletion is not a steady-state condition. If an MOS capacitor is held under such bias conditions, its capacitance will gradually increase toward C_{min} as the thermally generated minority charge builds up in the inversion layer until an equilibrium state is established. The time it takes for an MOS capacitor to recover from deep depletion and return to equilibrium is referred to as the *retention time*. It is a good indicator of the defect density in the silicon wafer and is often used to qualify processing tools in a facility.

It is possible to obtain *low-frequency-like* C–V curves at high measurement frequencies. One way is to expose the MOS capacitor to intense illumination, which generates a large number of minority carriers in the silicon. Another commonly used technique is to form an n^+ region adjacent to the MOS device and connect it electrically to the p-type substrate (Grove, 1967). The n^+ region then acts like a reservoir of electrons which can exchange minority carriers freely with the inversion layer. In other words, the n^+ region is connected to the *surface channel* of the inverted MOS device. This structure is similar to that of a gated diode, to be discussed in Section 2.3.5. Based on the equivalent circuit in Fig. 2.37(b), the total MOS capacitance per unit area is given by

$$C_g = \frac{C_{ox}(C_d + C_i)}{C_{ox} + C_d + C_i}. \tag{2.205}$$

When the MOS device is biased well into strong inversion, the *inversion-layer capacitance* C_i can be approximated by

$$C_i = \frac{d(-Q_i)}{d\psi_s} = \frac{|Q_i|}{2kT/q} \tag{2.206}$$

using Eq. (2.192).

The majority and minority carrier contributions to the total capacitance can be separately measured in a *split* C–V setup shown in Fig. 2.39(a) (Sodini *et al.*, 1982).

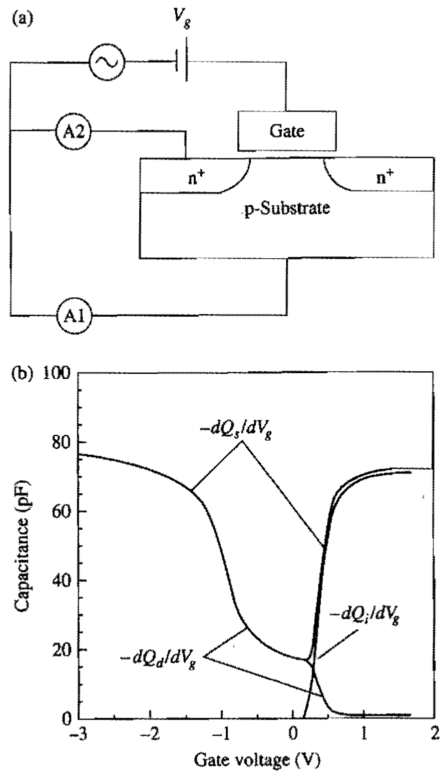


Figure 2.39. (a) Setup of the split C–V measurement. Both the dc bias and the small-signal ac voltage are applied to the gate. Small signal ac currents are measured by two ammeters, A1 and A2, connected separately as shown. (b) Measured C–V curves where the $-dQ_d/dV_g$ component is obtained from A1, and the $-dQ_i/dV_g$ component is obtained from A2. The sum is the total capacitance per unit area, $-dQ_s/dV_g$.

With a small signal ac voltage applied to the gate, the out-of-phase ac currents are sensed by two ammeters: one (A1) connected to the p-type substrate for the hole current, and another (A2) connected to the n^+ region for the electron current. Typical measured results are shown in Fig. 2.39(b). The hole contribution to the capacitance measured by A1 is

$$-\frac{dQ_d}{dV_g} = \frac{C_{ox}C_d}{C_{ox} + C_d + C_i}. \tag{2.207}$$

And the electron contribution to the capacitance measured by A2 is

$$-\frac{dQ_i}{dV_g} = \frac{C_{ox}C_i}{C_{ox} + C_d + C_i}. \tag{2.208}$$

They add up to the total capacitance per unit area, $C_g = -dQ_s/dV_g$. Note that the $-dQ_d/dV_g$ curve decreases to zero soon after strong inversion when C_i (Eq. (2.206)) becomes

dominant ($\gg C_{ox}$). To put it in another way, the highly conductive inversion channel shields the majority carriers in the bulk silicon so they do not respond to the modulation of gate field. The $-dQ_i/dV_g$ curve can be integrated to yield the inversion charge density as a function of the gate voltage. It is used, for example, in channel mobility measurements where the inversion charge density must be determined accurately.

2.3.4 Polysilicon-Gate Work Function and Depletion Effects

2.3.4.1 Work Function and Flatband Voltage of Polysilicon Gates

In the mainstream CMOS VLSI technology thus far, n^+-polysilicon gate has been used for nMOSFET and p^+-polysilicon gate for pMOSFET to obtain threshold voltages of low magnitude in both devices. The Fermi level of heavily doped n^+ polysilicon is near the conduction band edge, so its work function is given by the electron affinity, $q\chi$. From Eq. (2.169), the work function difference for an n^+ polysilicon gate on a p-type substrate of doping concentration N_a is

→ como si fuera un metal.

$$\phi_{ms} = -\frac{E_g}{2q} - \psi_B = -0.56 - \frac{kT}{q}\ln\left(\frac{N_a}{n_i}\right) \tag{2.209}$$

in volts. Similarly, the work function difference for a p^+ polysilicon gate on an n-type substrate of doping concentration N_d is

$$\phi_{ms} = \frac{E_g}{2q} + \psi_B = 0.56 + \frac{kT}{q}\ln\left(\frac{N_d}{n_i}\right), \tag{2.210}$$

which is symmetric to Eq. (2.209). These relations give rise to flatband voltages with key implications on the scalability of MOSFET devices, as will be discussed in Chapter 4.

The band diagram of an n^+-polysilicon-gated p-type MOS capacitor at zero gate voltage is shown in Fig. 2.40(a), where the Fermi levels line up and the free electron level of the bulk p-type silicon is higher in electron energy than the free electron level of the n^+ polysilicon gate. This sets up an oxide field in the direction of accelerating electrons toward the gate, and at the same time a downward bending of the silicon bands (depletion) toward the surface to produce a field in the same direction. The flatband condition is reached by applying a negative voltage equal to the work function difference to the gate, as shown in Fig. 2.40(b).

Problemas si el poly no está degenerado

2.3.4.2 Polysilicon-Gate Depletion Effects

The use of polysilicon gates is a key advance in modern CMOS technology, since it allows the source and drain regions to be self-aligned to the gate, thus eliminating parasitics from overlay errors (Kerwin *et al.*, 1969). However, if the polysilicon gate is not doped heavily enough, problems can arise from depletion of the gate itself. This is especially a concern with the dual n^+–p^+polysilicon-gate process in which the gates are doped by ion implantation (Wong *et al.*, 1988). Gate depletion results in an additional capacitance in series with the oxide capacitance, which in turn leads to a reduced inversion-layer charge density and degradation of the MOSFET transconductance.

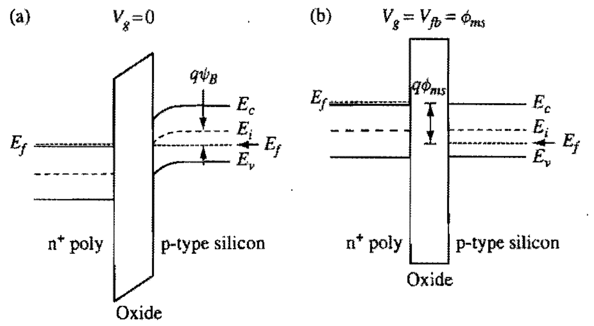


Figure 2.40. Band diagram of an n$^+$-polysilicon-gated p-type MOS capacitor biased at (a) zero gate voltage and (b) flatband condition.

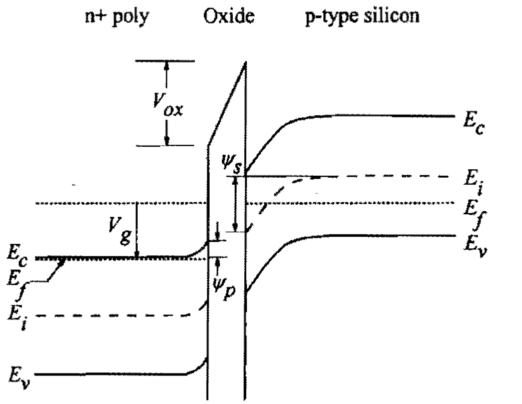


Figure 2.41. Band and potential diagram showing polysilicon-gate depletion effects when a positive voltage is applied to the n$^+$ polysilicon gate of a p-type MOS capacitor.

The analysis of MOS capacitance in the last subsection can be extended to include the polysilicon depletion effect and quantify certain observed features in the C–V characteristics. Consider the band diagram of an n$^+$-polysilicon-gated p-type MOS capacitor biased into inversion as shown in Fig. 2.41. Since the oxide field points in the direction of accelerating a negative charge toward the gate, the bands in the n$^+$ polysilicon bend slightly upward toward the oxide interface. This depletes the surface of electrons and forms a thin space-charge region in the polysilicon layer, which lowers the total capacitance.

2.3.4.3 Effect of Polysilicon Doping Concentration on C–V Characteristics

Typical low-frequency C–V curves in the presence of gate depletion effects are shown in Fig. 2.42 (Rios and Arora, 1994). A distinct feature is that the capacitance at inversion does not return to the full oxide capacitance as in Fig. 2.38. Instead, the inversion capacitance exhibits a maximum value somewhat less than C_{ox}, depending on the

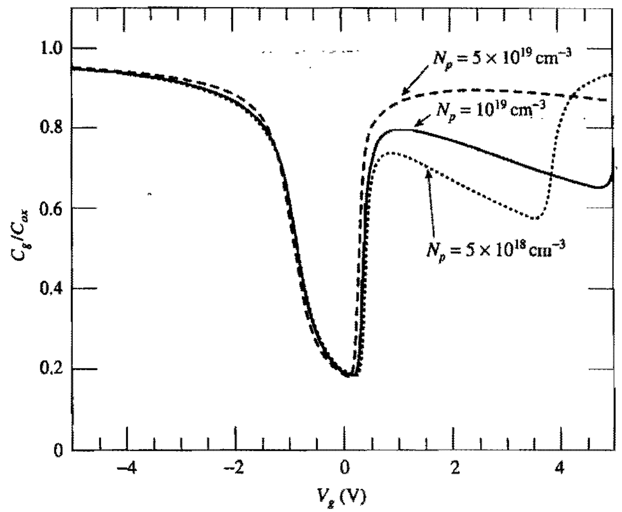


Figure 2.42. Low-frequency C–V curves of a p-type MOS capacitor with n$^+$ polysilicon gate doped at several different concentrations. (After Rios and Arora, 1994.)

effective doping concentration of the polysilicon gate. The higher the doping concentration, the less the gate depletion effect is and the closer the maximum capacitance is to the oxide capacitance.

The existence of a local maximum in the low-frequency C–V curve can be understood semiquantitatively as follows. In Fig. 2.41, we assume ψ_s to be the amount of band bending in the bulk silicon and ψ_p to be that in the n$^+$polysilicon. From charge neutrality, the total charge density Q_g of the ionized donors in the depletion region of the n$^+$ polysilicon gate is equal and opposite to the combined inversion and depletion charge density Q_s in the silicon substrate, i.e., $Q_g = -Q_s$. The gate bias equation for an applied voltage V_g is obtained by adding an additional term, ψ_p, for the band bending in the polysilicon gate to Eq. (2.195):

$$V_g = V_{fb} + \psi_s + \psi_p - \frac{Q_s}{C_{ox}}. \tag{2.211}$$

Differentiating Eq. (2.211) with respect to $-Q_s$ and using the capacitance definitions (2.196) and (2.197), we obtain

$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{1}{C_{si}} + \frac{1}{C_p}, \tag{2.212}$$

where $C_p = -dQ_s/d\psi_p = dQ_g/d\psi_p$ is the capacitance of the polysilicon depletion region.

When a p-type MOS device is biased well into strong inversion such that Q_s is dominated by the inversion charge, the low-frequency capacitance is approximately given by Eq. (2.206), $C_{si} = |Q_s|/(2kT/q) = Q_g/(2kT/q)$. Based on the depletion approximation, Eqs. (2.189) and (2.201), the polysilicon depletion capacitance can be expressed in terms of the depletion charge density Q_g as $C_p = \varepsilon_{si}qN_p/Q_g$, where N_p is the doping concentration of the polysilicon gate. Substituting these expressions into Eq. (2.212) yields

$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{2kT/q}{Q_g} + \frac{Q_g}{\varepsilon_{si}qN_p}. \tag{2.213}$$

As V_g becomes more positive, C_{si} ($\propto Q_g$) increases but C_p ($\propto 1/Q_g$) decreases. This results in a local maximum of the low-frequency capacitance as observed in Fig. 2.42. For $N_p < 10^{19}$ cm^{-3}, another abrupt rise in the MOS capacitance may be observed at a much higher gate voltage, as shown in Fig. 2.42. This is due to the onset of inversion (to p$^+$) at the n$^+$ polysilicon surface.

In practice, polysilicon gates cannot be doped much higher than 1×10^{20} cm^{-3}. Modern CMOS devices often operate at a maximum oxide field of 5 MV/cm, corresponding to a sheet electron density of $Q_g/q = 10^{13}$ cm^{-2}. For these values, the polysilicon depletion capacitance has the equivalent effect as adding 3–4 Å to the gate oxide thickness. The above discussion applies to p$^+$ polysilicon gates on n-type silicon as well as to n$^+$ polysilicon gates on p-type silicon. Note that for n$^+$ polysilicon gates on n-type silicon and p$^+$ polysilicon gates on p-type silicon, gate depletion occurs when the substrate is accumulated.

2.3.5 MOS under Nonequilibrium and Gated Diodes

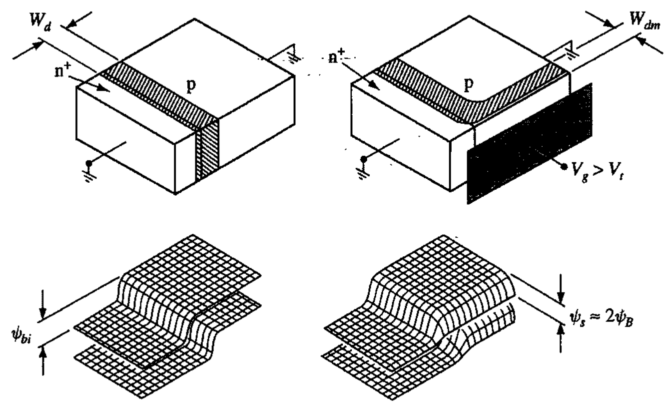
An important building block of VLSI devices is the gated diode, or gate-controlled diode. Consider an MOS capacitor where there is an n$^+$ region adjacent to the gated p-type region (Grove, 1967). The n$^+$ region and the p-type region form an n$^+$–p diode. This structure is shown schematically in Fig. 2.43 and was mentioned briefly at the end of Section 2.3.3.

2.3.5.1 Inversion Condition of an MOS Under Nonequilibrium

As discussed in Section 2.2.1, when both the n$^+$ region and the p-type substrate are connected to the same potential (grounded), the p–n junction is in equilibrium and the Fermi level is constant across the p–n junction. If the gate voltage is large enough to invert the p-type surface, which occurs for a surface potential bending of $\psi_s(\text{inv}) = 2\psi_B$, the inverted channel is connected to the n$^+$ region and has the same potential as the n$^+$ region. In other words, ***the electron quasi-Fermi level in the channel region is the same as the Fermi level in the n$^+$ region*** as well as the Fermi level in the p-type substrate. The depletion region now extends from the p–n junction to the region under the gate between the inverted channel and the substrate, as shown in Fig. 2.43(b).

If, on the other hand, the p–n junction is reverse-biased at a voltage V_R as shown in Fig. 2.44, the MOS is in a nonequilibrium condition in which $np \neq n_i^2$. From Eq. (2.107), the electron concentration on the p-type side of the junction is

$$n = \frac{n_i^2}{N_a}e^{-qV_R/kT}, \tag{2.214}$$



(a) (b)

Figure 2.43. Gated diode or p-type MOS with adjacent n$^+$ region in equilibrium (zero voltage across the p–n junction). The gate is biased at (a) flatband and (b) inversion conditions. (After Grove, 1967.)

since the Fermi level of the n$^+$ region is now qV_R lower than the Fermi level in the p-type substrate.[7] Now consider the case when a positive voltage large enough to bend the bands by $2\psi_B$ is applied to the gate, as shown in Fig. 2.44(b). This brings the conduction band at the surface $2\psi_B$ closer to the electron quasi-Fermi level. From Eq. (2.178), the electron concentration at the surface is increased by a factor of $\exp(2q\psi_B / kT) = (N_a/n_i)^2$ over that of Eq. (2.214), i.e.,

$$n = \frac{n_i^2}{N_a}e^{2q\psi_B/kT}e^{-qV_R/kT} = N_a e^{-qV_R/kT}. \tag{2.215}$$

Since this is much lower than the depletion charge density N_a, the surface remains depleted. Even though the positive voltage is sufficient to invert the surface in the equilibrium case, it is not enough to cause inversion in the reverse-biased case. This is because ***the reverse bias lowers the quasi-Fermi level of electrons so that even if the bands at the surface are bent as much as in the equilibrium case in Fig. 2.43(b), the conduction band is still not close enough to the quasi-Fermi level of electrons for inversion to occur.***

To reach inversion in the nonequilibrium case, a much larger gate voltage, sufficient to bend the bands by $2\psi_B + V_R$, must be applied. This is the case shown in Fig. 2.44(c), where the electron concentration at the surface is now

$$n = \frac{n_i^2}{N_a}e^{q(2\psi_B+V_R)/kT}e^{-qV_R/kT} = N_a, \tag{2.216}$$

the same as the condition for inversion introduced in Section 2.3.2. Notice that the surface depletion layer is much wider than in the equilibrium case, just as in a reverse-biased p–n junction.

[7] While this is not exactly true, it will not affect later results. Further discussions follow Eq. (2.216).

Posta q' no se entiende nada

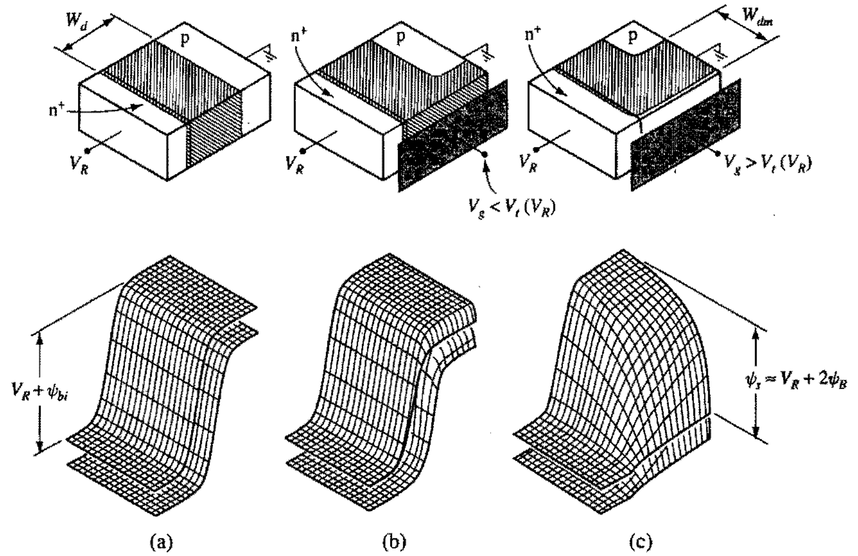


Figure 2.44. Gated diode or p-type MOS with adjacent n^+ region under nonequilibrium (reverse bias across the p–n junction). The gate is biased at (a) flat-band, (b) depletion, and (c) inversion conditions. (After Grove, 1967.)

To be more exact, Eq. (2.107), on which Eq. (2.214) is based, is not valid under a large reverse bias V_R. It is discussed in Appendix 4 that at large reverse biases, the electron quasi-Fermi level on the p-type side of the depletion region (at $x = 0$ in Fig. A4.5) is higher than the Fermi level of the n-type region (to the left of $x = -W_d$). In other words, the electron quasi-Fermi level on the p-type side of the depletion region is displaced downward from the Fermi level of the p-type region by less than qV_R. While this is true at the flatband condition in the gated diode depicted in Fig. 2.44(a), once a positive gate voltage is applied to bend down the p-type bands as in Fig. 2.44(c), the electron concentration at the p-type surface increases and the electron quasi-Fermi level there moves down toward the Fermi level of the n-type region. As far as the bands and the electron concentration at the p-type surface are concerned, the positive gate voltage has a similar effect as moving from the p-type side of the depletion region (at $x = 0$) toward the n-type side of the depletion region (at $x = -W_d$) in Fig. A4.5. When the threshold condition is reached, the electron quasi-Fermi level at the p-type surface would be the same as the Fermi level of the n-type region, i.e., displaced downward from the Fermi level of the p-type region by exactly qV_R. Another way to see this is that, in a region where the electron concentration is appreciable, electron quasi-Fermi level must remain essentially flat since the leakage current in a reverse-biased diode, $J_n \propto n\, d\phi_n/dx$, is negligibly small. In any case, the band bending requirement for threshold depicted in Fig. 2.44(c), $\psi_s = V_R + 2\psi_B$, is always valid.[8]

[8] This condition applies to the neutral p-type region. Less band bending is needed to reach the threshold for a point at the surface but within the depletion region of the reverse biased p–n junction, as is evident in Fig. 2.44(c).

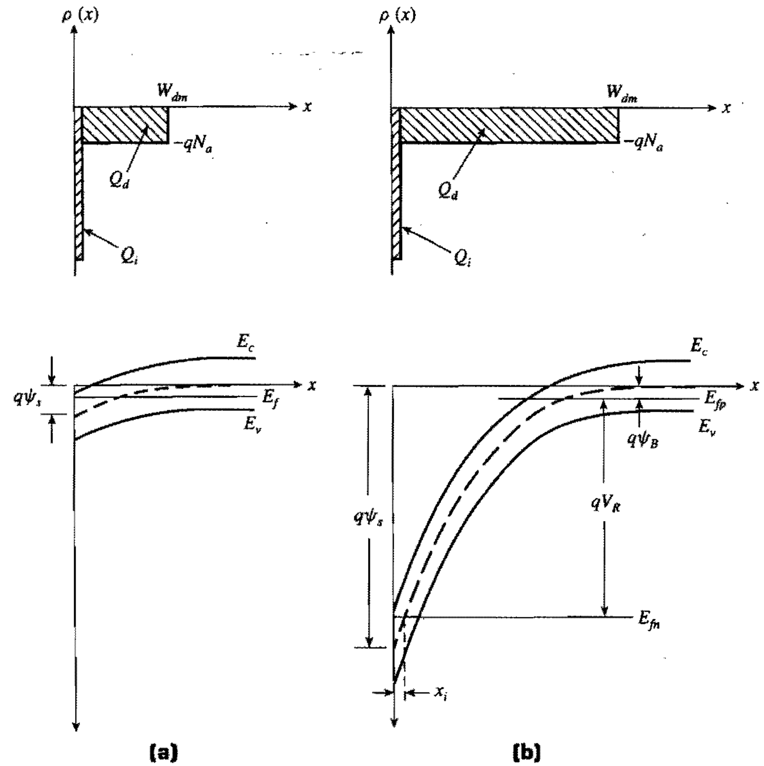


Figure 2.45. Comparison of charge distribution and energy-band variation of an inverted p-type region for (a) the equillibrium case and (b) the nonequillibrium case. (After Grove, 1967.)

2.3.5.2 Band Bending and Charge Distribution of an MOS Under Nonequilibrium

The above discussions are further illustrated in Fig. 2.45, where the charge distribution and band bending in a cross section perpendicular to the gate through the neutral p-type region are shown for both the equilibrium and the nonequilibrium cases. The equilibrium case is the same as that discussed in Section 2.3.2. In the nonequilibrium case, the hole quasi-Fermi level is the same as the Fermi level in the bulk p-type silicon, but the electron quasi-Fermi level is dictated by the Fermi level in the n^+ region (not shown in Fig. 2.45), which is qV_R lower than the p-type Fermi level. As a result, surface inversion occurs at a band bending

$$\psi_s(\text{inv}) = V_R + 2\psi_B, \tag{2.217}$$

and the maximum depletion width is a function of the reverse bias V_R,

$$W_{dm} = \sqrt{\frac{2\varepsilon_{si}(V_R + 2\psi_B)}{qN_a}}, \tag{2.218}$$

from Eq. (2.188).

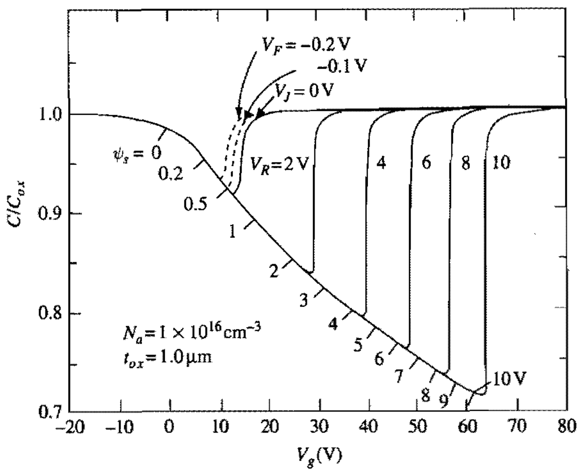


Figure 2.46. Normalized C–V characteristics of a p-type MOS capacitor with an adjacent n^+ region as that shown in Fig. 2.44(c). V_g is the voltage applied to the gate with respect to the p-type substrate. A series of C–V curves are shown for a range of V_R: the reverse bias voltage applied to the n^+/p junction.

When the surface is depleted, the gated diode behaves like an n^+–p diode with depletion of the p-region extending to underneath the gate electrode. When the surface is inverted, the gated diode behaves like an n^+–p diode with both the n^+ region and depletion of the p-region extending to underneath the gate electrode. High-field effects in gated diodes are discussed in Section 2.5.5

For a p-type MOS capacitor, the effect of an adjacent reverse biased n^+ region on the C–V characteristics is shown in Fig. 2.46. With increasing V_g, the surface potential ψ_s also increases, as labeled under the curve. At $V_R = 0$, the C–V curve resembles a regular low-frequency C–V curve (curve (a) in Fig. 2.38) with inversion (sharp rise of the capacitance to C_{ox}) taking place at $V_g \approx 13$ V where $\psi_s \approx 0.7$ V $(2\psi_B)$. Note that as V_R increases, the onset of inversion shifts to increasingly more positive gate voltages as the MOS goes into deeper and deeper depletion (lower C_{min}). It can be seen that the value of surface potential at inversion increases by approximately V_R, consistent with Eq. (2.217). The decrease of C_{min}, the serial combination of C_{ox} and $C_d = \varepsilon_{si}/W_{dm}$, with V_R follows from Eq. (2.218).

2.3.6 Charge in Silicon Dioxide and at the Silicon–Oxide Interface

It is often said that the real magic in silicon technology lies not in the silicon crystalline material but in silicon dioxide. Silicon dioxide forms critical components of silicon devices, serves as insulation and passivation layers, and is often used as an effective masking and/or diffusion-barrier layer in device fabrication.

Thus far we have treated silicon dioxide as an ideal insulator, with no space charge in or associated with it, and no charge exchange between it and the silicon it covers. The

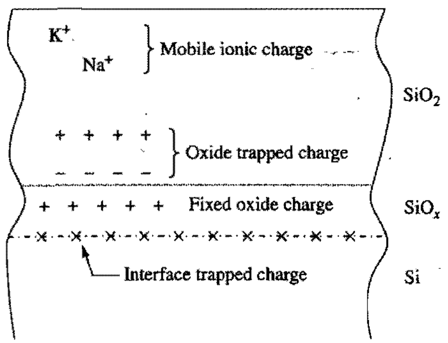


Figure 2.47. Charges and their location in thermally oxidized silicon. (After Deal, 1980.)

silicon dioxide and the oxide–silicon interface in real devices are never completely electrically neutral. There can be mobile ionic charges, electrons, or holes trapped in the oxide layer. There can also be fabrication-process-induced fixed oxide charges near the oxide–silicon interface, and charges trapped at the so-called *surface states* at the oxide–silicon interface. Electrons and holes can make transitions from the crystalline states near the oxide–silicon interface to the surface states, and vice versa. Since every device has some regions that are covered by silicon dioxide, the electrical characteristics of a device are very sensitive to the density and properties of the charges inside its oxide regions and at its silicon–oxide interface.

The nomenclature for describing the charges associated with the silicon dioxide in real devices was standardized in 1978 (Deal, 1980). The net charge per unit area is denoted by Q. Thus, Q_m denotes the *mobile charge* per unit area, Q_{ot} denotes the *oxide trapped charge* per unit area, Q_f denotes the *fixed oxide charge* per unit area, and Q_{it} denotes the *interface trapped charge* per unit area. The names and locations of these charges are illustrated in Fig. 2.47. The properties and characteristics of these charges are discussed further below.

2.3.6.1 Surface States and Interface Trapped Charge

At the Si–SiO_2 interface, the lattice of bulk silicon and all the properties associated with its periodicity terminate. As a result, localized states with energy in the forbidden energy gap of silicon are introduced at or very near the Si–SiO_2 interface (Many *et al.*, 1965). These localized surface states are illustrated schematically in Fig. 2.48. Interface trapped charges are electrons or holes trapped in these states.

Just like impurity energy levels in bulk silicon discussed in Section 2.1.2, the probability of occupation of a surface state by an electron or by a hole is determined by the surface-state energy relative to the Fermi level. Thus, as the surface potential is changed, the energy level of a surface state, which is fixed relative to the energy-band edges at the surface, moves with it. This change relative to the Fermi level causes a change in the probability of occupation of the surface state by an electron. For instance, referring to Fig. 2.32, as the bands are bent downward, or as the surface potential is increased,

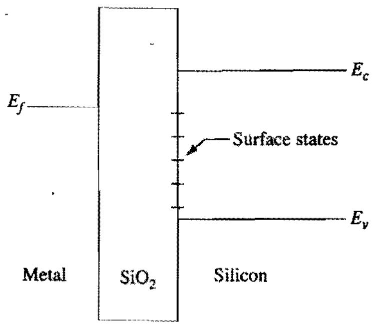


Figure 2.48. Schematic energy-band diagram of an MOS structure, illustrating the presence of surface states.

more surface states move below the Fermi level and hence become occupied by electrons. This change of interface trapped charge with a change in the surface potential gives rise to an additional silicon capacitance component, which will be discussed further in Section 2.3.7.

Electrons in silicon but near an oxide–silicon interface can make transitions between the conduction-band states and the surface states. An electron in the conduction band can contribute readily to electrical conduction current, while an electron in a surface state, i.e., an interface trapped electron, does not contribute readily to electrical conduction current, except by *hopping* among the surface states or by first making a transition to the conduction band. Similarly, holes in silicon but near an oxide–silicon interface can make transitions between the valence-band states and the surface states, and trapped interface holes do not contribute readily to electrical conduction. By trapping electrons and holes, surface states can reduce the conduction current in MOSFETs. Furthermore, the trapped electrons and holes can act like charged scattering centers, located at the interface, for the mobile carriers in a surface channel, and thus lower their mobility (Sah *et al.*, 1972).

Surface states can also act like localized generation–recombination centers. Depending on the surface potential, a surface state can first capture an electron from the conduction band, or a hole from the valence band. This captured electron can subsequently recombine with a hole from the valence band, or the captured hole can recombine with an electron from the conduction band. In this way, the surface state acts like a recombination center. Similarly, a surface state can act like a generation center by first emitting an electron followed by emitting a hole, or by first emitting a hole followed by emitting an electron. Thus, the presence of surface states can lead to surface generation–recombination leakage currents.

The density of surface states, and hence the density of interface traps, is a function of silicon substrate orientation and a strong function of the device fabrication process (EMIS, 1988; Razouk and Deal, 1979). In general, for a given device fabrication process, the dependence of the interface trap density on substrate orientation is $\langle 100\rangle < \langle 110\rangle < \langle 111\rangle$. Also, a postmetallization or "final" anneal in hydrogen, or in a hydrogen-containing ambient, at temperatures around 400 °C is quite effective in minimizing the density of interface traps. Consequently, $\langle 100\rangle$ silicon and postmetallization anneal in hydrogen are commonly used in modern VLSI device fabrication.

2.3.6.2 Fixed Oxide Charge

Fixed oxide charges are positive charges located in the oxide layer very close to the Si–SiO_2 interface. In fact, for modeling purposes, the fixed oxide charges are usually assumed to be located at the Si–SiO_2 interface. They are primarily due to excess silicon species introduced during oxidation and during postoxidation heat treatment (Deal *et al.*, 1967). The dependence of the density of fixed oxide charges on substrate orientation is the same as that of interface traps, namely $\langle 100\rangle < \langle 110\rangle < \langle 111\rangle$.

The presence of fixed oxide charges at the oxide–silicon interface affects the potential in the silicon, which will be discussed in the next subsection. In addition, the fixed oxide charges act as charged scattering centers and thus reduce the mobility of the carriers in a surface inversion channel (Sah *et al.*, 1972).

2.3.6.3 Mobile Ionic Charge

Mobile ionic charges in SiO_2 are usually due to sodium or potassium contamination introduced during device fabrication. Unlike fixed oxide charges, which are not mobile, Na^+ and K^+ ions are quite mobile in SiO_2 and can be moved from one end of the oxide layer to the other when an electric field is applied across the oxide layer, particularly at somewhat elevated temperatures (>200 °C) (Hillen and Verwey, 1986). As these positively charged ions drift close to the Si–SiO_2 interface, they repel holes from, and attract electrons to, the silicon surface, often causing unwanted surface electron current to flow among n^+ diffusion regions in a p-type substrate or well. Also, when these positively charged ions come close to the silicon surface, they can act as charged scattering centers for the carriers in the surface inversion channel, thus reducing their mobility.

In VLSI fabrication processes, ***mobile-ion contamination problems must be avoided.*** This is accomplished by a combination of proper passivation, usually using phosphosilicate glass, and "clean" fabrication technology (Hillen and Verwey, 1986).

2.3.6.4 Oxide Trapped Charge

If electron–hole pairs are generated in an oxide layer, e.g., by ionizing radiation, some of these electrons and holes can be subsequently trapped in the oxide. Also, if electrons or holes are injected into an oxide layer, by tunneling or by hot-carrier injection, some of them can be trapped in the oxide.

Electron and hole traps in SiO_2 can easily be introduced by bombardment with high-energy photons or particles (Bourgoin, 1989). Since bombardment by high-energy particles and photons is involved in many steps in the fabrication of modern VLSI devices (during ion implantation, plasma or reactive-ion etching, sputtering deposition, electron-beam evaporation of metal, electron-beam and x-ray lithography, etc.), electron and hole traps are often introduced in the oxide during device fabrication. Fortunately, most of these traps can be eliminated with subsequent anneals at temperatures above 550 °C (Ning, 1978).

Also, depending on the oxidation condition, electron traps can be introduced during the oxide growth process itself (EMIS, 1988). For example, oxide growth in moisture-containing ambient is known to introduce electron traps (Nicollian *et al.*, 1971).

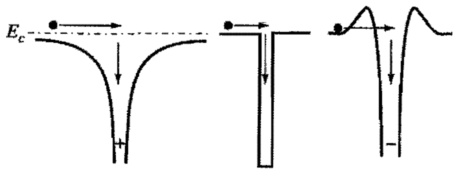


Figure 2.49. Schematics illustrating the potential wells of electron traps in silicon dioxide: (a) Coulomb-attractive trap, (b) neutral trap, and (c) Coulomb-repulsive trap.

- *Capture cross section.* Traps are usually characterized by their *capture cross sections.* Electron traps with cross sections in the range of 10^{-14}–10^{-12} cm^2 are usually *Coulomb-attractive* traps, i.e., the trap centers are positively charged prior to electron capture (Ning *et al.*, 1975; Lax, 1960). Electron traps with cross sections in the 10^{-18}–10^{-14}-cm^2 range are usually due to neutral traps (Lax, 1960), and those with cross sections smaller than 10^{-18} cm^2 are usually associated with *Coulomb-repulsive* traps, i.e., the trap centers are already negatively charged prior to electron capture (Balland and Barbottin, 1989). The potential wells representing these electron traps are illustrated in Fig. 2.49. Since the Coulomb-attractive and neutral centers have the largest capture cross sections, they are also the most important to include when considering the effects of electron traps on device characteristics. Hole traps have not been studied in as much detail as electron traps. This may be due to the fact that holes are very readily trapped when they are injected into an oxide layer (Goodman, 1966). This is consistent with the measured hole capture cross section of about 3×10^{-13} cm^2, which is as large as the largest electron traps in SiO_2 (Ning, 1976a).
- *Temperature dependence.* Consider the capture of a mobile electron into an electron trap. The trapping process has two competing components, namely, capturing the electron into some initial high-energy state of the trap center, and reemitting that same electron from the initial captured state by thermal excitation. If an electron in an initial captured state has a higher probability of cascading down towards the ground state of the trap center than of being reemitted, the electron becomes trapped. On the other hand, if the probability of reemission by thermal excitation from the initial captured state is high enough, trapping will not occur (Lax, 1960). The capture cross section, therefore, decreases with increasing temperature, since the probability for thermal reemission increases with temperature (Lax, 1960).
- *Field dependence.* If an electric field is applied across an oxide layer, it has the effect of increasing the energy of the carriers moving in the oxide layer. As these carriers gain energy from the oxide field, the probability of their being captured in some initial trap state is lowered, since the carriers now must lose more energy in the initial capture process. At the same time, an oxide field has the effect of lowering the energy barriers for the carriers trapped in a potential well, thus increasing the probability for reemitting them from their initial captured states (Lax, 1960). As a result, the capture cross section decreases with increasing oxide field (Ning, 1976b, 1978). ***The commonly used method of injecting carriers into SiO_2 by tunneling at high oxide fields tends to underestimate the amount of traps*** in SiO_2, since the capture cross sections at such high oxide fields are much smaller than those at normal device operation.

2.3.7 Effect of Interface Traps and Oxide Charge on Device Characteristics

The presence of oxide charges and interface traps has three major effects on the characteristics of devices. First, the charge in the oxide, or in the interface traps, interacts with the charge in the silicon near the surface and thus changes the silicon charge distribution and the surface potential. Second, as the density of interface trapped charge changes with changes in the surface potential, it gives rise to an additional capacitance component in parallel with the silicon capacitance C_{si} discussed in Section 2.3.3. Third, the interface traps can act as generation–recombination centers, or assist in the band-to-band tunneling process, and thus contribute to the leakage current in a gated-diode structure. These effects are discussed more quantitatively below.

2.3.7.1 Effect of Oxide Charge on Surface Potential

As discussed in Section 2.3.2, the charge distribution in silicon is a function of the surface potential. Thus, the effect of oxide charge on the charge distribution in silicon can be described in terms of its effect on surface potential. In the case of an MOS structure, the effect of oxide charge is usually described in terms of the change in gate voltage, which is a readily measurable parameter, necessary to counter the effect of the oxide charge or to restore the surface potential to that of zero oxide charge.

For simplicity of illustration, let us consider an MOS structure biased at flat-band condition. Let us assume that a sheet of oxide charge Q per unit area is placed at a distance x from the gate electrode, and a gate voltage δV_g has been applied to restore the MOS structure to its original, i.e., flat-band, condition. With the surface potential restored to its original value, the sheet of oxide charge has induced no change in the charge distribution in the silicon, which is a function of the surface potential, but a charge of magnitude $-Q$ per unit area on the gate electrode. This is illustrated in Fig. 2.50. Gauss's law in Eq. (2.43) implies that the electric field in the oxide between 0 and x due to the sheet of oxide charge and its image charge on the gate electrode is $-Q/\varepsilon_{ox}$ (see Exercise 2.5). This is also illustrated in Fig. 2.50. The potential difference supporting this electric field is $-xQ/\varepsilon_{ox}$, which is provided by the applied gate voltage. Therefore,

$$\delta V_g = -\frac{xQ}{\varepsilon_{ox}}. \tag{2.219}$$

The gate voltage necessary to offset the effect of an arbitrary oxide charge distribution can be obtained by superposition of individual elements of the charge distribution and applying Eq. (2.219) to each element. For an oxide charge distribution of $\rho_{net}(x, \psi_s) = \rho_{net}(x) + Q_{it}(\psi_s)\delta(x - t_{ox})$, which consists of an arbitrary distribution $\rho_{net}(x)$ that is independent of the surface potential and a delta-function distribution of the interface trap charge $Q_{it}(\psi_s)$ located at $x = t_{ox}$, the gate voltage necessary to offset it is

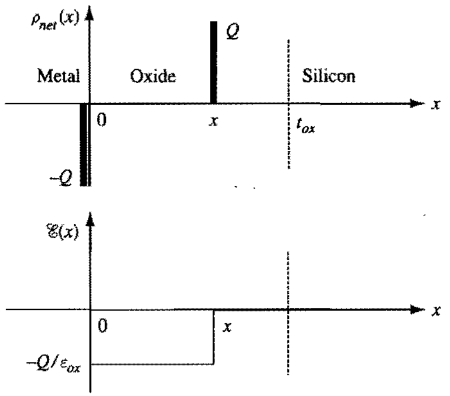


Figure 2.50. Schematic illustrating the effect of a sheet charge of areal density Q within the oxide layer of an MOS capacitor biased at flat-band condition.

$$
\begin{aligned}
\Delta V_g = \Delta V_g(\psi_s) &= -\frac{1}{\varepsilon_{ox}} \int_0^{t_{ox}} x\rho_{net}(x, \psi_s)dx \\
&= -\frac{1}{\varepsilon_{ox}} \left(\int_0^{t_{ox}} x\rho_{net}(x)dx + Q_{it}(\psi_s)t_{ox} \right).
\end{aligned} \tag{2.220}
$$

According to the charge nomenclature discussed in Subsection 2.3.6, $\rho_{net}(x)$ includes the mobile charge, the oxide trapped charge, and the fixed oxide charge. It is a common practice to define an *equivalent oxide charge per unit area*, Q_{ox}, by

$$
\begin{aligned}
Q_{ox} = Q_{ox}(\psi_s) &\equiv \int_0^{t_{ox}} \frac{x}{t_{ox}} \rho_{net}(x, \psi_s)dx \\
&= \int_0^{t_{ox}} \frac{x}{t_{ox}} \rho_{net}(x)dx + Q_{it}(\psi_s).
\end{aligned} \tag{2.221}
$$

Equation (2.220) can then be rewritten in the simple form of

$$
\Delta V_g(\psi_s) = -\frac{Q_{ox}(\psi_s)}{C_{ox}}, \tag{2.222}
$$

where $C_{ox} = \varepsilon_{ox}/t_{ox}$ is the oxide capacitance per unit area introduced in Eq. (2.195). Equation (2.222) states that the effect of an arbitrary oxide charge distribution is equivalent to an oxide sheet charge of areal density $Q_{ox}(\psi_s)$ located at the oxide–silicon interface.

2.3.7.2 Interface-Trap Capacitance

In Section 2.3.3, the silicon part of the capacitance is defined without including any interface trapped charge. As the interface traps are filled and emptied in response to changes in the surface potential, they give rise to an *interface-trap capacitance per unit area*, C_{it}, defined by

$$
C_{it}(\psi_s) \equiv -\frac{dQ_{it}(\psi_s)}{d\psi_s}. \tag{2.223}
$$

Just as in Eq. (2.197), the − sign in Eq. (2.223) is inserted to ensure that the capacitance is always a positive quantity. In Eq. (2.223), we have indicated explicitly that the interface-trap capacitance is a function of surface potential.

To include the effect of Q_{ox} in the operation of an MOS capacitor, Eq. (2.195) should be modified by adding to its right-hand side a ΔV_g term due to Q_{ox}. That is, Eq. (2.195) becomes

$$
\begin{aligned}
V_g - V_{fb} &= \Delta V_g(\psi_s) - \frac{Q_s(\psi_s)}{C_{ox}} + \psi_s \\
&= -\frac{Q_s(\psi_s) + Q_{ox}(\psi_s)}{C_{ox}} + \psi_s.
\end{aligned} \tag{2.224}
$$

The total charge on the gate electrode is now $Q_s(\psi_s) + Q_{ox}(\psi_s)$. Equation (2.196) then becomes

$$
C_g = -\frac{d[Q_s(\psi_s) + Q_{ox}(\psi_s)]}{dV_g}, \tag{2.225}
$$

and Eq. (2.198) becomes

$$
\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{1}{C_{si} + C_{it}}. \tag{2.226}
$$

That is, the interface-trap capacitance is in parallel with the silicon capacitance.

As discussed in the previous subsection, the probability of a surface state being filled with an electron is governed by its energy level relative to the Fermi level. ***Only those interface traps that can be filled and emptied at a rate faster than the capacitance-measurement signal can contribute to C_{it}.*** Traps too slow to follow the capacitance-measurement signal will not contribute to C_{it}.

2.3.7.3 C–V Curves as a Monitoring and Diagnostic Tool for Oxide and Interface Quality

As discussed in Section 2.3.6, the amount of oxide charge and surface states is a function of the silicon substrate orientation and a strong function of the device fabrication process. For modern MOS devices, by the time a device fabrication process is ready for manufacturing, the amount of oxide charge and surface states is usually quite low, with Q_{ox}/q typically about 10^{11} cm^{-2} or less. For an MOS device having an oxide thickness of 10 nm, the corresponding gate voltage shift according to Eq. (2.222) is only 46 mV. At such low surface-state densities, the MOS C–V characteristics are quite ideal in that the measured C–V curves match well with the calculated ones (see Section 2.3.3 and Fig. 2.38).

However, many experimental fabrication processes, particularly those involving high-energy plasma, reactive-ion etching, and electron or ion beams, can generate significant oxide charge and surface states in MOS devices. Unless these *oxide damages* can be removed by post-process thermal annealing, the amount of residual oxide charge and surface states can be appreciable. High-field stress of silicon MOS devices can also

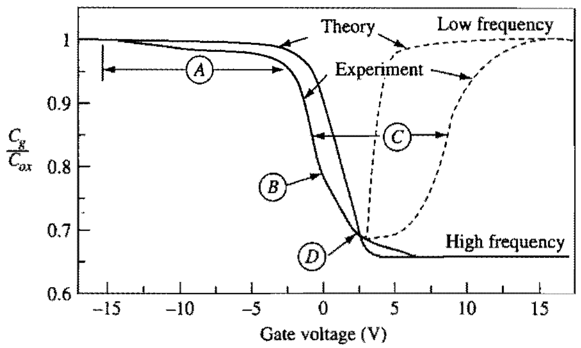


Figure 2.51. Comparison of experimental and theoretical high-frequency and low-frequency C–V curves, showing typical distortion caused by interface traps. The MOS capacitor has $N_a = 10^{16}$ cm^{-3} and $t_{ox} = 200$ nm. The symbols are explained in the text. (After Deal *et al.*, 1969.)

generate oxide charge and surface states. (High-field effects will be discussed in Section 2.5.) The presence of oxide charge and surface states can cause the measured C–V curve to appear distorted compared to the ideal C–V curve. There are two contributions to this distortion. First, Q_{it} is a part of Q_{ox} which shifts the C–V curve along the gate voltage axis according to Eq. (2.224). The shift is distorted because Q_{it} is a function of surface potential, which in turn is a function of gate voltage. Second, the additional capacitance due to the interface traps also distorts the measured C–V curve because C_{it} is a function of surface potential, which in turn is a function of gate voltage.

If the amount of oxide charge and surface states is large, the distortions in the C–V curve can be quite prominent, as illustrated in Fig. 2.51 (Deal *et al.*, 1969). The physical mechanisms responsible for the various distorted regions can be understood as follows. The distortion labeled A is where the MOS capacitor is normally in accumulation. In this gate voltage region, the valence-band edge at the silicon–oxide interface approaches or crosses the Fermi level (see Fig. 2.31). As a result, the interface states near the valence band become ionized and positively charged. (The interface states near the valence band are called donor states. They are neutral when they lie below the Fermi level and become positively charged by donating electrons when they lie above the Fermi level.) As the donor interface states become ionized, they contribute to a build up of positive interface trap charge which shifts the gate voltage in the negative direction according to Eq. (2.224). The distortion near the label B is related to interface states near the midgap, since it occurs at a gate voltage range where the MOS capacitor is between flatband and weak-inversion conditions (see Figs 2.31 and 2.38). The distortion labeled D is where the MOS capacitor is near weak inversion. To the right side of D, the capacitor is in inversion where the conduction-band edge at the silicon–oxide interface approaches or crosses the Fermi level. In this gate voltage range, the interface states near the conduction band become ionized and negatively charged. (The interface states near the conduction band are called acceptor states. They are neutral when they lie above the Fermi level and become negatively charged by accepting electrons when they lie below the Fermi level.) As the acceptor interface states become ionized, they contribute to a build up of negative interface trap charge which shifts

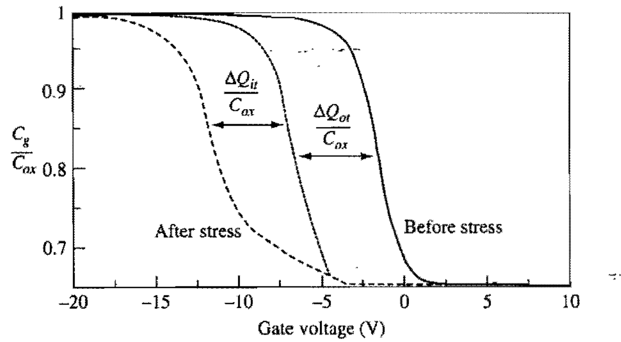


Figure 2.52. Typical high-frequency C–V plot of an MOS capacitor showing the distortion due to interface traps. The MOS capacitor has $N_a = 10^{16}$ cm^{-3} and $t_{ox} = 200$ nm. The oxide trapped charge and interface trapped charge are caused by subjecting the capacitor to a negative bias stress of 2MV/cm at 400 °C for 2 minutes. (After Deal *et al.*, 1967.)

the gate voltage in the positive direction according to Eq. (2.224). This causes the low-frequency C–V curve to shift to the right. The broadening of the C–V curve at its midpoint is labeled C. It is a result of the interface states near the conduction band (Deal *et al.*, 1969).

Figure 2.52 illustrates the distortion of a typical high-frequency C–V curve of an MOS capacitor after trapped charge has been created inside the oxide layer and at the oxide–silicon interface (Deal *et al.*, 1967). (The creation of bulk oxide and interface traps by high electric fields will be discussed in Section 2.5.) The oxide trapped charge causes a parallel shift of the C–V curve (dotted line) to the left. The interface-trap capacitance causes the curve to be distorted and shifted to the left by an additional amount.

The C–V distortions depicted in Figs 2.51 and 2.52 are for 200 nm thick oxides having significant oxide charge and surface states. It can be inferred from Eqs. (2.221) and (2.222) that the magnitude of the gate voltage shift caused by a certain areal density of interface states, Q_{it}, is proportional to t_{ox}. The voltage shift caused by a certain uniform volume density of oxide trapped charge, ρ_{net}, is proportional to t_{ox}^2. Therefore, for the same Q_{it} and ρ_{net}, the C–V curves of thinner oxide devices should appear less distorted.

There is a vast amount of published literature on the subject of interface states and the measurement of interface states. Interested readers are referred to the literature for a discussion on the general characteristics of interface states in MOS capacitors (Deal *et al.*, 1969) and on the various techniques for measuring interface states (Schroder, 1990).

2.3.7.4 Surface Generation–Recombination Centers

As discussed in the previous subsection, interface states can serve as generation–recombination centers. In the case of a gated-diode structure, the surface generation–recombination current adds to the diode leakage current. The magnitude of the surface leakage current depends on whether or not the surface states are *exposed*, i.e., whether or not the silicon surface is depleted (Grove and Fitzgerald, 1966). If the surface is inverted, the surface states are all filled with minority carriers and do not function efficiently as generation centers. Similarly, if the surface is in accumulation, the surface states are all filled with majority carriers and do not function efficiently as generation centers either. ***Only when the silicon surface is depleted will the surface states function efficiently as generation centers.*** Thus, surface leakage current can be suppressed by biasing the gate to keep the silicon surface either in inversion or in accumulation. The reader is referred to Appendix 5 for a detailed discussion of the physics involved in generation–recombination processes.

As recombination centers, surface states can degrade the minority-carrier lifetime of devices. Consequently, devices where long minority-carrier lifetimes are required are usually designed to confine the minority carriers in them away from the silicon surface. In addition, the device fabrication processes are usually optimized to minimize the density of surface states.

2.3.7.5 Surface-State or Trap-Asssisted Band-to-Band Tunneling

As will be discussed in Subsection 2.5.2, band-to-band tunneling occurs when the electric field across a p–n junction is sufficiently large. For a gated diode, or for a p–n diode with silicon surface components, the presence of surface states or interface traps in the high-field region can enhance the band-to-band tunneling current very significantly. Thus, gate-induced drain leakage currents in MOSFETs, which will be discussed in Section 2.5.5, and emitter–base diode tunneling currents in bipolar transistors, which will be discussed in Section 6.3.4, depend strongly on the density of interface states at the oxide–silicon interface of these devices.

2.4 Metal–Silicon Contacts

The metal–semiconductor contact is a critically important element in all semiconductor devices and technology. As a contact to a silicon device terminal, a metal–silicon contact should be non-rectifying and have a small contact resistance in order to minimize the voltage drop across the contact. Such contacts are usually referred to as *ohmic contacts*. In general, a metal–semiconductor contact has rectifying current–voltage characteristics similar to those of a p–n diode (see Section 2.2). Rectifying metal–semiconductor devices are called *Schottky diodes* or *Schottky barrier diodes*. Here we discuss the basic physics and operation of a metal–silicon contact, focusing on its current–voltage characteristics as a Schottky diode and as an ohmic contact. A brief discussion of Schottky diodes as active devices is also given.

2.4.1 Static Characteristics of a Schottky Barrier Diode

The static characteristics of a Schottky barrier diode can be inferred from those of an MOS capacitor (see Section 2.3) by letting the oxide layer thickness go to zero. Just as the surface potential of the semiconductor in an MOS capacitor is affected by the interface trapped charge, the surface potential of the semiconductor in a Schottky barrier diode is affected by the electron occupation of the surface states on the semiconductor surface. Therefore, the characteristics of a Schottky barrier diode depend on the properties of the metal and the properties of the semiconductor and its surface states. Here we first discuss

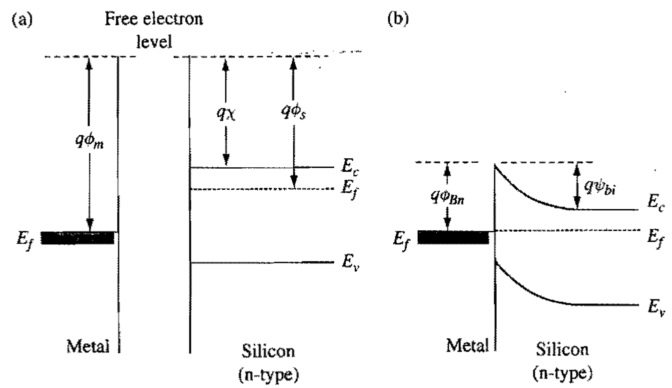


Figure 2.53. Energy-band diagrams of a metal–silicon system where the silicon surface is assumed to be absent of any surface states. (a) When the metal and the silicon are far apart. (b) When the metal is in contact with the silicon, with no externally applied voltage.

the static characteristics of a Schottky barrier diode ignoring all surface states, and then discuss how the surface states can modify the diode characteristics.

2.4.1.1 Schottky Barrier Diodes Without Surface States

When surface states are ignored, the energy-band diagram of an MOS capacitor shown in Fig. 2.29 can be readily adapted to give the energy-band diagram of a metal–silicon system. It was pointed out in Section 2.3.1.1 that when two different materials are brought into contact, they must share the same free electron level at the interface. Also, it was shown in Sections 2.1.1.3 and 2.1.4.5 that at thermal equilibrium or when there is no net electron or hole current through a system, the Fermi level of the system is spatially constant. These two factors together lead to the energy-band diagrams in Fig. 2.53 for a metal–n-silicon contact at thermal equilibrium. From consideration of free electron level at the interface, the Schottky barrier height for electrons, $q\phi_{Bn}$, is

$$q\phi_{Bn} = q(\phi_m - \chi), \tag{2.227}$$

where $q\phi_m$ is the metal work function and $q\chi$ is the electron affinity of silicon. From consideration of Fermi level being spatially constant, the built-in potential ψ_{bi} is

$$\begin{aligned} q\psi_{bi} &= q\phi_{bn} - (E_c - E_f)_{bulk} \\ &= q\left(\phi_{bn} - \frac{E_g}{2q} + \psi_B\right), \end{aligned} \tag{2.228}$$

where $\psi_B = |\psi_f - \psi_i|$ [see Eq. (2.48)].

The built-in potential implies an amount of depletion-layer charge Q_d in the silicon given by Eq. (2.189). To maintain overall charge neutrality of the metal–silicon system, this depletion charge induces a sheet of electronic charge of the same density per unit area in the metal at the metal–silicon interface.